

Quantum Braid Dynamics

A Computational Process

R. Fisher

May 31, 2026

Abstract

This monograph presents Quantum Braid Dynamics (QBD), a novel computational framework that derives the continuous fabric of spacetime and the foundations of quantum mechanics from discrete, background-independent causal graph dynamics. We formulate the ontological substrate of the universe as a dynamic causal graph governed by strict topological and algebraic constraints, including a dual logical-physical time architecture, strict irreflexivity, and acyclicity. By establishing a stabilizer codespace structure over causal diamonds, we construct a fault-tolerant topological quantum error-correcting code inherent to the pre-geometric vacuum, where physical updates correspond to logical operations. We define the dynamic evolution of the system through a comonadic self-observation and stochastic rewrite constructor, calibrating fundamental constants such as the vacuum temperature and self-energy from information-theoretic principles. Finally, we prove the geometric well-posedness and convergence of the discrete graph sequence to a smooth, globally hyperbolic four-dimensional Lorentzian manifold under the Lorentzian Gromov-Hausdorff-Prokhorov metric. This framework unifies general relativistic geometry and quantum fault tolerance, establishing that the laws of physics emerge naturally from the thermodynamic limits of discrete information processing.

Chapter 1: Substrate (Ontology)

The Foundational Principles

The Rules

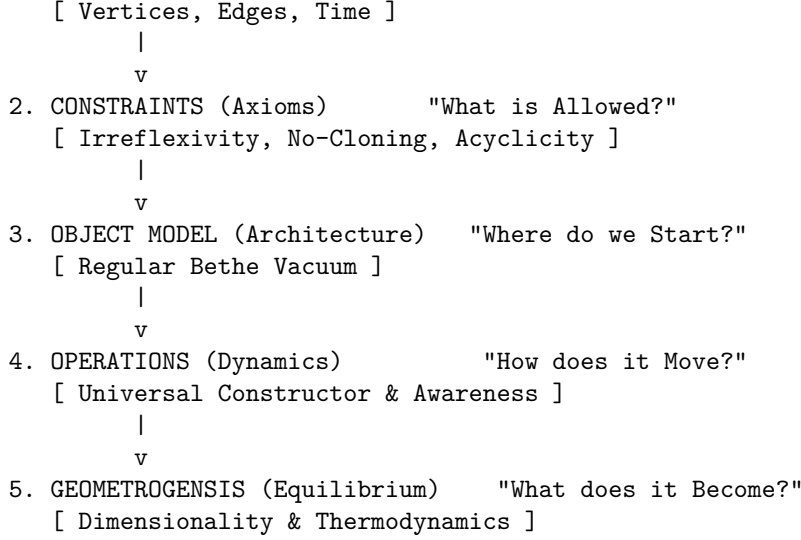
Quantum Braid Dynamics, *A Computational Process* (QBD) is presented in a form explicitly engineered for auditability. This format ensures ideas become pure logic that can be parsed, producing a physical theory that is unambiguous and well defined, whose meaning is fully determined by its internal logic.

In Part 1, The Foundational Principles begins the construction of the physical universe as a deductive chain, moving from abstract requirements to concrete emergence. Chapter 1 defines the minimal substrate of existence. Strict axiomatic constraints enforce causality and prevent logical paradoxes in Chapter 2, distinguishing the physically possible from the mathematically constructible. The unique initial state of the universe is revealed in Chapter 3 as a topological structure poised for evolution which is animated by a dynamical engine in Chapter 4, a universal constructor driven by information-theoretic potentials that dictate how connections evolve. Finally, Chapter 5 demonstrates how the collective action of this process yields a stable macroscopic phase of spacetime through thermodynamic equilibrium, bridging discrete graph dynamics and continuous geometry.

PART 1: THE FOUNDATIONAL PRINCIPLES (The Rules)

=====

1. SUBSTRATE (Ontology) "What Exists?"



Chapter 1: Substrate (Ontology)

Standard physics typically grants itself a pre-existing stage: a manifold of space and time where particles interact and fields propagate. Yet we find that this assumption obscures the very origin of the structure we wish to understand. To derive the architecture of the universe, our shared inquiry cannot start by assuming the building already exists. It becomes necessary to descend to a level more primitive than geometry, seeking a substrate that possesses neither location nor duration until those properties are constructed. We must ask how a universe can exist before there is a “where” for it to exist in, or a “when” for it to happen.

Discarding the continuum, with its implication of infinite information density within every volume, reveals itself as a logical necessity if we are to respect the limits of computation and the finiteness of information. A domain of pure relation remains for us to analyze. We must strip away the comfortable illusions of smooth space to find the discrete gears operating beneath the fabric. If we do not, we trap ourselves in the paradoxes of the infinite before we have even begun to describe the finite.

The task at hand involves understanding how independent, dimensionless events can weave themselves into a sequence that mimics the flow of time and a network that mimics the extension of space. We must determine how a collection of dimensionless points can develop the properties of adjacency, distance, and direction without referencing an external coordinate system. This chapter on Ontology establishes the epistemological rules that permit us to build such a model, defines the discrete nature of the temporal iterator that drives it, and constructs the relational graph that serves as the absolute floor of our reality.

Preconditions and Goals

- Establish unprovability of axioms to justify a coherentist epistemological approach.
 - Define dual-time architecture separating logical iteration (t_L) from emergent physical time (t_{phys}).
 - Construct the causal graph $G = (V, E, H)$ as the fundamental substrate of existence.
 - Derive necessity of a finite temporal origin to prevent infinite regress paradoxes.
 - Formalize the symmetry of the Elementary Task Space to ensure kinematic neutrality.
-

1.1 Epistemological Foundations

A logical hazard confronts us immediately when we attempt to define the absolute bottom of reality. This is the ancient problem of the infinite regress of justification. If we demand that our foundation be rigorously

proven, we are compelled to provide a set of prior axioms to construct that proof. Those prior axioms, in turn, require their own antecedents to validate them, and so we fall into a bottomless well of requirements. It becomes clear that a physical theory cannot claim absolute security if its roots cannot be proven from within its own system. However, logic dictates that no system can prove its own consistency without stepping outside of itself. We must therefore shift our standard of validity entirely. We cannot demand that our axioms be self-evident truths handed down from above. Instead, we must select them as consistent and fertile tools that justify themselves solely by the universe they are capable of building. We are looking for a starting point that does not require an antecedent.

We must look to the structure of deductive systems to understand where the limits of certainty lie. Standard approaches in physics often attempt to anchor their theories in self-evident truths or undeniable observations. However, Gödel teaches us that in any sufficiently powerful formal system, there are truths that cannot be proven syntactically. If we persist in searching for a pre-written scroll of absolute truth that requires no justification, we trap ourselves in a state of intellectual paralysis. We are not uncovering an archaeological artifact that was hidden in the sand. We are designing a machine of logic that must run without crashing. This realization frees us from the impossible demand of absolute certainty and allows us to focus on the engineering constraint of structural coherence. Our goal is not to find the one true axiom but to find an axiom that works.

1.1.1 Unprovability of Axioms

Inherent Unprovability of Axiomatic Foundations arising from the Structure of Deductive Systems

It is established as a structural necessity of deductive logic that within any finite formal system $\mathfrak{S}$, the chain of justification for any proposition p must terminate in a set of foundational propositions, designated as the **Axiomatic Basis** ($\mathcal{A}$), for which no antecedent justification exists within $\mathfrak{S}$. The truth value of any element $a \in \mathcal{A}$ is determined by postulate, not by syntactic derivation from a prior theorem. Consequently, the concept of proving an axiom within the system it generates constitutes a logical contradiction, as any such proof would require the axiom to serve as its own premise or derive from a circular chain, both of which invalidate the proof structure.

The enterprise of deductive reasoning, the bedrock of mathematics and logic, is built upon a foundational paradox. Any attempt to establish an ultimate truth through proof must contend with the Munchhausen trilemma: the chain of justification must either regress infinitely, loop back upon itself in a circle, or terminate in a set of propositions that are accepted without proof. In the architecture of formal deductive systems, these terminal propositions are known as axioms. Historically, they were considered self-evident truths, but modern logic has recast them as foundational assumptions. A distinction is made between a syntactic process of derivation from accepted premises and a justification, which is the meta-systemic, philosophical, and pragmatic argument for adopting those premises in the first place.

A foundational axiomatic structure is a coherent set of postulates whose justification rests not on derivational dependency or claims of self-evidence, but on the systemic utility and coherence of the entire theoretical edifice it supports. The selection of axioms is a rational process motivated by criteria such as parsimony, consistency, and the richness of the consequences (the theorems) that can be derived from them. This perspective on selection is, therefore, a conclusion forced by the evolution of mathematics itself. The historical journey from a classical view of axioms as immutable truths to a modern, formalist view of axioms as definitional starting points reflects a profound epistemological shift. This transition, catalyzed by the discovery of non-Euclidean geometries, revealed that the “truth” of an axiom lies not in its correspondence to a singular, external reality, but in its role in defining a consistent and fruitful logical system.

To build this argument, the formal definitions that govern deductive systems are first established, then the logical necessity of unprovable truths is explored through the lens of Gödel’s incompleteness theorems. Subsequently, two pivotal case studies from the history of mathematics are analyzed: the centuries-long debate over Euclid’s parallel postulate and the more recent controversy surrounding the Axiom of Choice.

These examples are framed within a coherentist epistemology, distinguishing this holistic mode of justification from fallacious circular reasoning. Finally, an analogy is drawn to the foundational postulates of Relational Quantum Mechanics to demonstrate the broad applicability of this justificatory framework across the formal and physical sciences.

+-----+		
	THE MUNCHHAUSEN TRILEMMA	
	(The Three Failures of Absolute Justification)	
+-----+		
1. INFINITE REGRESS (Ad Infinitum)		
+-----+		
	A <- justified by B <- justified by C...	
+-----+		
2. CIRCULARITY (Petitio Principii)		
+-----+		
	A <- justified by B <- justified by A	
+-----+		
3. AXIOMATIC STOPPING (Dogmatism)		
+-----+		
	A <- justified by "Self-Evidence"	
	(The "Foundational Cut")	
+-----+		

1.1.2 Deductive System Components

Definition of Formal Deductive Systems as a Tripartite Framework of Language, Axioms, and Inference

A **Formal Deductive System** is defined as the tripartite structure $\mathfrak{D} = (\mathcal{L}, \mathcal{A}, \mathcal{I})$, comprising the following immutable components:

1. $\mathcal{L}$, the **Formal Language**, consisting of a finite alphabet and a recursive grammar that defines the set of all possible Well-Formed Formulas (WFFs).
2. $\mathcal{A}$, the **Axiomatic Basis**, a distinct, finite subset of formulas accepted as premises without internal proof.
3. $\mathcal{I}$, the set of **Rules of Inference**, defining computable transformations that map a finite set of premises to a valid conclusion.

A **Proof** is strictly defined as a finite sequence of formulas wherein each member is either an element of $\mathcal{A}$ or derived directly from preceding members via the application of $\mathcal{I}$.

To comprehend the distinction between proof and justification, the precise structure of the environment in which proofs exist must first be understood. A formal, or deductive, system is an abstract framework composed of three essential components: a formal language; a set of axioms; a set of rules of inference.

The formal language consists of an alphabet of symbols and a grammar that specifies how to construct well-formed formulas (WFFs), which are the legitimate statements of the system. The axioms and rules of inference constitute the “rules of the game,” defining how these statements can be manipulated.

Axioms: Logical vs. Non-Logical

Axioms themselves are divided into two categories:

- **Logical axioms:** Statements that are considered universally true within the framework of logic itself, often taking the form of tautologies. An example is the schema $(A \supset B) \supset A$, which holds regardless of the specific content of propositions A and B . These axioms are foundational to reasoning in any domain.
- **Non-logical axioms** (also known as postulates or proper axioms): Substantive assertions that define a particular theory or domain of inquiry, such as geometry or set theory. The statement $a + 0 = a$ is a non-logical axiom defining a property of integer arithmetic.

The Nature of Formal Proof

Within this defined system, a formal proof is a finite sequence of WFFs where each statement in the sequence is either:

- an axiom;
- a pre-stated assumption; or
- derived from preceding statements in the sequence by applying a rule of inference.

The final statement in the sequence is called a theorem. This definition is critical because it structurally separates axioms from theorems. Axioms are, by definition, the statements that begin a deductive chain; they cannot, therefore, be the conclusion of one (Enderton, 2001). The very structure of a formal system thus makes the concept of “proving an axiom” an internal contradiction.

A proof is a sequence $S_1, S_2, \dots, S_n$, where S_n is the theorem. Each S_i must be an axiom or follow from previous sentences via an inference rule. If an axiom A were to be proven, it would have to be the final sentence in such a sequence. But that sequence must start from other axioms. If it does, then A is not an axiom but a theorem derived from those other axioms. If the proof of A requires A itself as a premise, the reasoning is circular and thus not a valid proof. Consequently, within any non-circular, deductive system, axioms are definitionally unprovable.

Truth, Validity, Soundness, and Completeness

This syntactic process of derivation must be distinguished from the semantic concept of truth. Logicians differentiate between:

- Syntactic derivability
 - denoted by $\vdash$
- Semantic entailment or truth
 - denoted by $\models$

An argument is valid if, in every possible interpretation or “world” where its premises are true, its conclusion is also true.

A deductive system is said to be:

- **Sound** if it only proves valid arguments; if a statement is derivable from a set of axioms, it is also semantically entailed by them.
 - If $\Gamma \vdash \theta$, then $\Gamma \models \theta$
- **Complete** if it can prove every valid argument.
 - If $\Gamma \models \theta$, then $\Gamma \vdash \theta$

This distinction is paramount: axioms are the starting points for the syntactic game of proof. Their justification, however, is a meta-systemic and semantic consideration, concerning what kind of “world” or “model” the syntactic system describes, and whether that model is consistent, coherent, and useful.

1.1.3 Godelian Incompleteness

Limits of Provability and Consistency imposed by Sufficiently Powerful Formal Systems

Pursuant to the Incompleteness Theorems, for any consistent formal system $\mathfrak{F}$ capable of expressing primitive recursive arithmetic, there exists a statement $\mathcal{G}$ such that $\mathfrak{F} \not\vdash \mathcal{G}$ and $\mathfrak{F} \not\vdash \neg\mathcal{G}$. Furthermore, the consistency of the system itself, denoted $Con(\mathfrak{F})$, cannot be derived using the resources of $\mathfrak{F}$ alone. Therefore, the validity of the axiomatic foundation $\mathcal{A}$ cannot be established by the deductive machinery it enables; justification must be sought through meta-systemic criteria.

The unprovability of axioms, while definitionally true, was elevated from a structural feature to a fundamental law of logic by the work of Kurt Godel. Before Godel, one could still harbor the ambition, as exemplified by the logicist program of Gottlob Frege and Bertrand Russell, of reducing the vast edifice of mathematics to a minimal set of purely logical axioms. The goal was to show that mathematical truths were simply complex tautologies. Godel's incompleteness theorems demonstrated that this foundationalist dream was, for any sufficiently powerful system, mathematically impossible.

Godel's Incompleteness Theorems

In 1931, Godel published his two incompleteness theorems, which irrevocably altered the philosophy of mathematics. (Godel, 1931)

- **The First Incompleteness Theorem** states that for any consistent, effectively axiomatized formal system F that is powerful enough to express the basic arithmetic of natural numbers, there will always be statements in the language of F that are true but cannot be proven within F . Godel's proof was constructive: he showed how to create such a statement, often called the Godel sentence $\mathcal{G}$, which can be informally interpreted as, "This statement is not provable in system F . If F is consistent, then $\mathcal{G}$ must be true, yet unprovable within F ."
- **The Second Incompleteness Theorem** is a corollary of the first. It states that such a system F cannot prove its own consistency. The statement of consistency, $Con(F)$, is another example of a true but unprovable proposition within F .

Implications for Axioms

These theorems have profound implications for the nature of axioms. They show that the set of "true" arithmetical statements is larger than the set of "provable" statements for any given axiomatic system. This means that no single, finite set of axioms can ever be complete; there will always be mathematical truths that lie beyond its deductive reach. The selection of an axiom set is therefore not a matter of discovering the "one true" foundation, but rather a choice to explore the consequences of a particular set of assumptions, with the full knowledge that these assumptions will be inherently incomplete.

Furthermore, the Second Incompleteness Theorem shows that our confidence in the consistency of a foundational system like Zermelo-Fraenkel set theory (ZFC) cannot come from a proof within ZFC itself. This belief must be grounded in meta-systemic reasoning (such as the fact that no contradictions have been found after decades of intense scrutiny, or the construction of models in other theoretical frameworks). This is a form of justification, not a formal proof.

Godel's work transformed the status of axioms from potentially self-evident truths into necessary epistemic leaps. It proved that incompleteness is not a flaw to be fixed but a fundamental property of formal reasoning. This realization forces the justification of axioms away from the foundationalist hope of a complete, self-verifying system and toward a pragmatic, coherentist framework where axioms are judged by their power and consistency, not their claim to absolute, provable truth.

1.1.4 Euclidean Geometry

Shift from Self-Evidence to Consistency through the History of the Parallel Postulate

The history of Euclid's fifth postulate provides the quintessential example of the evolution in how axioms are justified. It marks the transition from a foundationalist appeal to self-evidence and correspondence with physical reality to a modern, coherentist justification based on internal consistency and systemic definition.

Euclid's Elements and the Ambiguous Fifth Postulate

In his *Elements*, Euclid established a system of geometry based on five postulates. The first four are simple, constructive, and intuitively appealing:

- A straight line can be drawn between any two points.
- A line segment can be extended indefinitely.
- A circle can be drawn with any center and radius.
- All right angles are equal.

The fifth postulate, however, is notably more complex. In its original form, it states that if two lines are intersected by a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines must intersect on that side if extended far enough. This statement, which is logically equivalent to the more familiar Playfair's axiom ("through a point not on a given line, there is exactly one line parallel to the given line"), felt less like a self-evident truth and more like a theorem in need of proof. Euclid's own apparent reluctance to use it until the 29th proposition of his work suggests he may have shared this view.

The Quest for a Proof (c. 300 BCE-1800 CE)

For over two millennia, mathematicians attempted to prove the fifth postulate from the first four. Figures from Ptolemy in antiquity to Arab mathematicians like Ibn al-Haytham and Omar Khayyam, and later European scholars like Girolamo Saccheri, dedicated themselves to this task. Each attempt ultimately failed. The invariable error was to unknowingly assume a hidden proposition that was itself logically equivalent to the parallel postulate. For instance, proofs would implicitly assume that the sum of the angles in a triangle is always 180 deg, or that similar triangles of different sizes exist: both of which are consequences of the fifth postulate, not the first four alone. These repeated failures were, in retrospect, powerful evidence for the postulate's independence from the others.

The Non-Euclidean Revolution

The decisive breakthrough came in the early 19th century with the work of Carl Friedrich Gauss, Janos Bolyai, and Nikolai Lobachevsky. Instead of trying to derive the fifth postulate, they boldly explored the consequences of negating it. By assuming that through a point not on a line there could be infinitely many parallel lines, they developed a completely new, logically consistent system: hyperbolic geometry. Similarly, the assumption that there are no parallel lines gives rise to elliptic geometry. These non-Euclidean geometries contained bizarre and counterintuitive theorems, such as triangles whose angles sum to less than 180 deg (hyperbolic) or more than 180 deg (elliptic), yet they were internally free of contradiction.

Justification Through Consistency: The Beltrami-Klein Model

The existence of these formal systems was not enough; their legitimacy required a demonstration of their consistency. This was definitively achieved by Eugenio Beltrami in the 1860s. Beltrami constructed a model of the hyperbolic plane within Euclidean space. In what is now known as the Beltrami-Klein model:

- the "plane" is the interior of a Euclidean disk;
- "points" are Euclidean points within that disk; and
- "lines" are the Euclidean chords of the disk.

Within this model, it is possible to demonstrate that all the axioms of hyperbolic geometry, including the negation of the parallel postulate, hold true. For any "line" (chord) and any "point" (internal point) not on it, one can draw infinitely many other "lines" (chords) through that point that do not intersect the first.

This model established the relative consistency of hyperbolic geometry: if Euclidean geometry is free from contradiction, then hyperbolic geometry must be as well. Any contradiction found in hyperbolic geometry could be translated, via the model, into a contradiction within Euclidean geometry. The justification for the axioms of hyperbolic geometry was therefore not an appeal to their “truth” about physical space, but a rigorous demonstration that they cohered into a consistent logical structure. This event fundamentally altered the understanding of axioms, shifting their role from describing a single reality to defining the rules for a multiplicity of possible, consistent worlds.

1.1.5 Axiom of Choice

Acceptance of Non-Constructive Principles based on Systemic Fertility

If the debate over the parallel postulate marked the birth of a new view on axioms, the controversy surrounding the Axiom of Choice represents its full maturation. Here, the justification for adopting a foundational principle is almost entirely divorced from physical intuition or self-evidence, resting instead on the internal coherence and sheer utility of the mathematical system it enables.

Introducing the Axiom of Choice

First formulated by Ernst Zermelo in 1904, the Axiom of Choice states that for any collection of non-empty sets, there exists a function (a “choice function”) that selects exactly one element from each set. For a finite collection, this is provable from more basic axioms. The power and controversy of AC arise when dealing with infinite collections. Bertrand Russell’s famous analogy clarifies its nature:

- Given an infinite collection of pairs of shoes, one can define a choice function (“for each pair, choose the left shoe”).
- But for an infinite collection of pairs of socks, where the two members of a pair are indistinguishable, no such defining rule exists.

AC asserts that a choice function nevertheless exists, even if it cannot be constructed or explicitly defined.

Controversy and Counterintuitive Consequences

This non-constructive character is the primary source of objection to AC, particularly from mathematicians of the constructivist and intuitionist schools, for whom “to exist” means “to be constructible”. The axiom’s acceptance leads to a number of deeply counterintuitive results that challenge our physical understanding. The most famous of these is the Banach-Tarski paradox, which demonstrates that a solid sphere can be decomposed into a finite number of non-overlapping pieces, which can then be reassembled by rigid motions to form two solid spheres, each identical in size to the original. This result appears to violate the conservation of volume, but the paradox is resolved by noting that the “pieces” involved are so complex that they are non-measurable, as they cannot be assigned a well-defined volume.

Justification through Systemic Utility and Equivalence

Despite these paradoxes, the Axiom of Choice is a standard and indispensable component of modern mathematics, forming the C in ZFC (Zermelo-Fraenkel set theory with Choice), the most common foundation for the field. Its justification is almost entirely pragmatic, stemming from its immense power and the elegance of the theories it facilitates. Within the context of the other ZF axioms, AC is logically equivalent to several other powerful and widely used principles, most notably:

- Zorn’s Lemma: This principle states that a partially ordered set in which every chain (totally ordered subset) has an upper bound must contain at least one maximal element.
- The Well-Ordering Principle: This principle asserts that any set can be “well-ordered,” meaning its elements can be arranged in an order such that every non-empty subset has a least element. These equivalent forms, particularly Zorn’s Lemma, are essential tools in numerous branches of mathematics. Their use is critical in proving fundamental theorems such as:
- Every vector space has a basis.
- Every commutative ring with a unit element contains a maximal ideal (Krull’s Theorem).

- The product of any collection of compact topological spaces is compact (Tychonoff's Theorem).

The mathematical community has largely accepted AC because rejecting it would mean abandoning these and countless other foundational results, effectively crippling vast areas of modern algebra, analysis, and topology. The justification is not its intuitive plausibility, but its mathematical fertility. The matter was settled formally when Kurt Godel (1938) and Paul Cohen (1963) proved that AC is independent of the other axioms of ZF set theory; it can be neither proved nor disproved from them. Its inclusion is a genuine choice, and that choice has been made in favor of systemic power over intuitive comfort (Marker, 2002).

1.1.6 Coherentist Justification

Justification of Unprovable Postulates by Coherentist Criteria

The historical evolution of axiomatic justification, as seen in the cases of the parallel postulate and the Axiom of Choice, points toward a specific epistemological framework: coherentism. This view contrasts sharply with the classical foundationalist approach that once dominated mathematical philosophy.

The justification for the adoption of the Axiomatic Basis $\mathcal{A}$ is determined exclusively by the **Coherence Criteria** of the generated system, defined as the conjunction of the following properties: 1. **Consistency**: The absolute inability to derive a contradiction ($\perp$) from $\mathcal{A}$. 2. **Independence**: The non-derivability of any axiom $a \in \mathcal{A}$ from the set difference $\mathcal{A} \setminus \{a\}$. 3. **Parsimony**: The minimization of the cardinality $|\mathcal{A}|$ relative to the explanatory power of the system. 4. **Fertility**: The capacity of the system to generate theorems that map isomorphically to observable physical phenomena.

Foundationalism vs. Coherentism in Epistemology

Foundationalism posits that knowledge is structured like a building, resting upon a secure foundation of basic, self-justifying beliefs. In mathematics, the classical view of axioms as “self-evident truth” is a quintessential form of foundationalism. These axioms were thought to be directly apprehended as true and required no further support; all other mathematical knowledge (theorems) was then built upon this unshakeable base.

In coherentism, the structure of knowledge is envisioned instead as Otto Neurath's famous ship, where each component is supported by its relationship to all the others within a holistic web of belief. The modern, formalist justification of axioms is explicitly coherentist. Axioms are chosen not because they are self-evident truths, but because they serve as the starting points for a system that, as a whole, exhibits desirable properties.

Criteria for a Coherent Axiomatic System

The justification for a set of axioms, from a coherentist perspective, is evaluated based on the properties of the entire system they generate. The primary criteria include:

- **Consistency**: The system must be free from internal contradiction. It should be impossible to derive both a proposition P and its negation $\neg P$ from the axioms. This is the absolute, non-negotiable requirement for any logical system.
- **Independence**: No axiom should be derivable from the others. While not strictly necessary for consistency, independence is highly valued according to the principle of parsimony, thus ensuring that the set of foundational assumptions is minimal.
- **Parsimony**: Often associated with Occam's Razor, this principle suggests that the set of axioms should be as small and conceptually simple as possible while still being sufficient to generate the desired theoretical framework.
- **Fertility (or Utility)**: The axiomatic system should be powerful and productive. It should generate a rich body of interesting and useful theorems, unify disparate results, and provide elegant proofs for known facts. This is the criterion that most strongly guided the acceptance of the Axiom of Choice.

Distinguishing Coherence from Fallacy (Petitio Principii)

A common objection to coherentism is that it endorses circular reasoning. However, there is a crucial distinction between the holistic justification of coherentism and the fallacy of *petitio principii*, or begging the question.

- **Petitio Principii:** This is a fallacy of linear argument where a conclusion is supported by a premise that is either identical to or already presupposes the conclusion. The argument “ P is true because P is true” provides no new support for P .
- **Coherentist Justification:** This is non-linear and holistic. An axiom A is not justified by an argument that presupposes A . Rather, A is justified because the entire system it generates (the set of axioms and all derivable theorems $\{A, T_1, T_2, \dots\}$) exhibits the virtues of consistency, parsimony, and fertility. The justification flows from the emergent properties of the whole system back to its foundational parts. The relationship is one of mutual support within an interconnected web, not a simple derivational loop.

Summary Table: Epistemological Approaches

Criterion	Foundationalist View (Classical)	Coherentist View (Modern/Formalist)
Nature of Axioms	Self-evident truths; descriptions of a pre-existing reality (mathematical or physical).	Foundational assumptions; definitions that construct a formal system.
Source of Justification	Direct intuition, self-evidence, correspondence to reality.	Systemic properties: consistency, parsimony, and the fertility/utility of the resulting theorems.
Structure of Knowledge	Linear and hierarchical. Theorems are built upon the unshakeable foundation of axioms.	Holistic and non-linear. Axioms and theorems are mutually supporting parts of a coherent web.
Response to Alternatives	Alternative axioms (e.g., non-Euclidean) are considered “false” as they do not correspond to reality.	Alternative axioms are valid starting points for different, equally consistent systems. The choice between them is pragmatic.

1.1.7 RQM Analogy

Relational Interpretation of Quantum Mechanics as an Epistemological Precedent

The model of coherentist justification for foundational postulates is not confined to pure mathematics. It finds a powerful parallel in the interpretation of fundamental physics, particularly in Carlo Rovelli’s Relational Quantum Mechanics (RQM). This interpretation offers a compelling case study of how choosing a new set of postulates, justified by their systemic coherence, can resolve long-standing conceptual problems.

Introduction to Relational Quantum Mechanics (RQM)

Proposed by Rovelli in 1996, RQM is an interpretation of quantum mechanics that challenges the notion of an absolute, observer-independent quantum state (Rovelli, 1996). The core tenet of RQM is that the properties of a physical system are relational; they are only meaningful with respect to another physical system (the “observer”). As Rovelli states, “different observers can give different accounts of the same set of events.”

Crucially, an “observer” in this context is not necessarily a conscious being but can be any physical system that interacts with another. A particle’s spin, for example, does not have an absolute value but only a value relative to the measuring apparatus that interacts with it.

The Foundational Postulates of RQM

Rovelli’s original formulation was motivated by information theory and based on two primary postulates:

1. There is a maximum amount of relevant information that can be extracted from a system (finiteness).
2. It is always possible to acquire new information about a system (novelty). More recent codifications of RQM list a set of principles, including:
 - Relative Facts: Events or facts occur relative to interacting physical systems.
 - No Hidden Variables: Standard quantum mechanics is complete.
 - Internally Consistent Descriptions: The descriptions from different observer perspectives, while different, must cohere in a predictable way when one observer measures another.

Justification of RQM's Postulates

These postulates are not justified because they are directly observable or self-evident. We cannot “see” the relational nature of a quantum state in an absolute sense. Instead, their justification is entirely coherentist and pragmatic. By adopting this relational framework, many of the most persistent paradoxes of quantum mechanics, such as the measurement problem (the “collapse of the wavefunction”) and the Schrodinger’s cat paradox, are removed without needing to invoke more radical physics, such as hidden variables (as in Bohmian mechanics) or a multiplicity of universes (as in the Many-Worlds Interpretation).

In RQM, the “collapse” is not a physical process happening in an absolute sense; it is simply the updating of an observer’s information about a system relative to their interaction. For a different observer who has not interacted with the system-observer pair, the pair remains in a superposition. The justification for RQM’s postulates is their explanatory power and their ability to create an internally consistent and coherent ontology for the quantum world, using only the existing mathematical formalism of the theory.

This process mirrors the justification of non-Euclidean geometry. The measurement problem in quantum mechanics played a role analogous to the problematic parallel postulate in geometry, an element that seemed at odds with the philosophical underpinnings of the rest of the theory. The solution was not to prove the old assumption (absolute state) but to replace it with a new one (relational states) and demonstrate that the resulting system is consistent and resolves the initial tension. In both mathematics and physics, the justification for a foundational leap lies in the coherence and problem-solving power of the new intellectual world it constructs.

1.1.8 Unprovability of Axioms

Formal Argument of Self-Validation Failure from the Structural Separation of Axioms and Theorems

This analysis has traced the distinction between the proof of a theorem and the justification of an axiom, arguing that the latter is a rational process grounded in systemic coherence and utility. The very definition of a formal deductive system renders its axioms unprovable from within; they are the starting points from which all proofs begin. Godel’s incompleteness theorems elevate this definitional truth to a fundamental proof, a limitation of logic, demonstrating that any sufficiently powerful axiomatic system is necessarily incomplete and cannot prove its own consistency. This mathematical reality precludes the foundationalist dream of a complete and self-verifying basis for all knowledge, forcing the acceptance of axioms to be an act of justified, meta-systemic choice.

The historical case studies of Euclidean geometry and the Axiom of Choice serve as powerful illustrations of this principle in action. The centuries-long effort to prove the parallel postulate gave way to the realization that it was an independent choice, defining one of several possible consistent geometries. Its justification shifted from an appeal to physical intuition to a demonstration of its role within a coherent system. The Axiom of Choice presents an even more modern case, where a physically counterintuitive and non-constructive principle is widely accepted based almost entirely on its mathematical fertility (the immense power and elegance of the theorems it makes provable).

This mode of justification is best understood through the epistemological framework of coherentism, where beliefs (or in this case, axioms) are validated by their mutual support within a larger system. This holistic process is distinct from fallacious circular reasoning. It is a rational, highly constrained procedure guided

by the principles of consistency, parsimony, and systemic utility. The analogy with Rovelli's Relational Quantum Mechanics underscores that this is not a feature unique to mathematics but a fundamental aspect of theory-building in the face of foundational questions.

Ultimately, foundational axioms are not the bedrock of truth in the sense of being immutable, provable facts. They are, rather, the architectural blueprints for vast and intricate systems of thought. An axiom is justified not because it is a self-evident point of departure, but because it is the cornerstone of a powerful, elegant, and coherent intellectual world. The act of justification is the demonstration that such a world can be built without collapsing into contradiction, and that the world so built is worth exploring.

1.1.Z Implications and Synthesis

Epistemological Foundations

We have justified our starting points by the physics they produce. This approach allows us to accept them without the impossible requirement of absolute, antecedent proof. By abandoning the search for a static or self-evident truth, we have committed to constructing logical self-consistency through a coherentist framework. We have traded the illusion of a proven foundation for the utility of a computable one. This clears the ground for a constructive physics that does not require an infinite chain of prior causes to function, it is a strategic alignment with the nature of formal systems. We acknowledge that the map must be drawn before it can be read.

This result reframes the role of the physicist from a discoverer of pre-existing laws to an architect of necessary logic. In a traditional reductionist view, one expects to find a bottom to reality in the form of particles or fields that simply exist without cause. However, the logic of deductive systems teaches us that any such foundation is arbitrary unless it justifies itself through operation. We are not digging for a foundation that sits passively beneath the universe. We are identifying the operating system that keeps the universe running. The truth of our axioms lies not in their divine origin but in their structural stability. We are asserting that the physical universe is isomorphic to a formal system because it is a deduction being executed. Therefore, the constraints we place upon our theory, such as finiteness and consistency, are ontological requirements for existence itself.

Furthermore, this finiteness imposes a strict boundary on the physical structure because it cannot support infinite histories or undefined origins. If the logic requires a starting point to be computable, we must conclude that the universe itself requires a starting moment. We cannot hide behind the concept of eternal cycles or infinite regress. These are computationally undefined operations that would prevent the system from ever initializing. This epistemological constraint forces our hand regarding the nature of time. The timeline cannot stretch back forever, or the logical system will fail to boot. We are thus compelled to construct a temporal ontology that respects these limits, leading us directly to the definition of the logical clock.

1.2 Temporal Ontology

Defining time in a universe that does not yet possess entropy or clocks presents a distinct challenge. While imagining a universal metronome ticking in the background is tempting, we know that in a background-independent theory, no such external reference exists. We must strip time down to its barest function. We must identify the mechanism that distinguishes one state from the next. Without this separation, there is no cause and effect. There is only a static singularity of information where everything happens at once. To rely on a pre-existing temporal coordinate would be to assume the very thing we are trying to derive. We must build time from the ground up as a process of change.

Distinguishing between the logical sequence of updates that drives the system and the physical time eventually

measured by observers within it becomes essential. We must separate the hand that moves the pieces from the experience of the pieces themselves. Ordering events requires a mechanism that operates independently of the geometry of spacetime because the assembly of spacetime has not yet occurred. We need a raw iterator. We need a counter that marks the progression of the computational process itself. This iterator acts as the heartbeat of the algorithm. It ensures that events occur in a definitive sequence even before the concept of duration exists.

Establishing a dual architecture for time resolves this difficulty by separating the iterator from the metric. We also confront the paradoxes inherent in an infinite past. If the universe had no beginning, the information required to describe the current state would be boundless. This would violate the finiteness criterion we just established. We demonstrate that the timeline must be bounded in the past to avoid physical contradictions like the Grim Reaper paradox. Therefore, we define a temporal domain that initiates at zero and advances by integer steps. This boundary condition is not merely a philosophical preference but a logical necessity for a constructive theory. A program cannot run if it never starts.

1.2.1 Postulate: Dual Time Architecture

Separation of Emergent Physical Time from Fundamental Logical Time through a Dual-Time Architecture

The temporal structure of the physical theory is constituted by two distinct, orthogonal, and non-interchangeable parameters: 1. **Global Logical Time (t_L)**: The fundamental ordering parameter of state evolution. The domain of t_L is strictly restricted to the set of non-negative integers $\mathbb{N}_0$. This parameter serves as the discrete iteration counter for the Universal Evolution Operator and is not subject to relativistic dilation or coordinate transformation. 2. **Physical Time (t_{phys})**: An emergent, continuous parameter derived from relational path lengths within the graph substrate. t_{phys} is subordinate to t_L and possesses geometric character, emerging only in the macroscopic limit.

The foundational postulate of this theory asserts that physical reality emerges as a secondary phenomenon rather than serving as a primary, self-subsistent entity; this assertion compels an immediate and total rupture with standard temporal formulations, thereby necessitating the complete rejection of all such formulations without any form of compromise or partial retention. In their place, the theory introduces a strict dual-time structure, wherein two distinct temporal parameters operate at orthogonal levels of ontological priority, each fulfilling precisely defined roles that preclude overlap or interchangeability.

This dual-time structure comprises the following two components, rigorously delineated to ensure no ambiguity arises in their application or interpretation:

- t_{phys} : This parameter emerges within the internal dynamics of the physical system itself; it is inherently relational, meaning its values derive solely from comparisons among events or states embedded within the system; it possesses a geometric character, aligning with the curved spacetime metrics of general relativity; it remains local in scope, applicable only to subsystems or observers confined to specific regions of the universe; it appears continuous in the effective macroscopic limit, where quantum discreteness averages out to yield smooth trajectories; and it becomes measurable exclusively through the agency of physical clocks, which are themselves constituents of the system and thus subject to the same emergent constraints.
- t_L : This parameter stands as the fundamental temporal scaffold upon which all physical emergence depends; it originates externally to the physical system, positioned at a meta-theoretical level that transcends the system's own dynamics; it manifests as strictly discrete, advancing only in integer increments without intermediate fractional values; it enforces an absolute ordering across the entirety of the universe's state sequence, providing a universal "before" and "after" that admits no exceptions or relativizations; it remains strictly unobservable from the vantage point of any internal state within the system, as no physical process can access or register its progression; and it functions solely as the iteration counter within the universal computation, tallying each discrete application of the evolution operator without contributing to the observable content of the states themselves.

This distinction between t_{phys} and t_L constitutes an indispensable structural necessity. It represents the sole known resolution capable of simultaneously accommodating the following five critical requirements of a viable physical theory:

1. Background independence, which demands that no fixed external arena preconditions the dynamics;
2. Finite information content, which prohibits unbounded informational resources at any finite stage;
3. Causal acyclicity, which ensures that the partial order of causation contains no closed loops;
4. Constructive definability, which mandates that all entities and processes arise from finite specifications;
5. The phenomenon of evolution, wherein states succeed one another and generate observable change.

Any attempt to merge or conflate these two temporal parameters would reintroduce at least one of the paradoxes afflicting prior formulations, such as the timeless stasis of the Wheeler-DeWitt constraint (Anderson, 2012).

1.2.2 Definition: Global Logical Time

Global Sequencer (t_L) as the Fundamental Iterator of State Evolution

$t_L \in \mathbb{N}_0$ constitutes the discrete, non-negative integer that systematically labels the successive global states of the universe as they arise under the repeated action of $\mathcal{U}$. Formally, this labeling traces the iterative progression of the universe's configuration through the following infinite but forward-directed chain:

$$U_0 \xrightarrow{\mathcal{U}} U_1 \xrightarrow{\mathcal{U}} U_2 \xrightarrow{\mathcal{U}} \dots \xrightarrow{\mathcal{U}} U_{t_L}$$

In this sequence, each application of $\mathcal{U}$ transforms the prior state U_{t_L} into the subsequent state U_{t_L+1} , preserving the necessary constraints while introducing the potential for structural evolution. t_L thereby imposes a strict total order on the entire sequence of states, establishing an unequivocal precedence relation such that for any $i < j$, the state U_i precedes U_j without ambiguity or overlap. Consequently, t_L emerges as the sole known parameter capable of distinguishing “before” from “after” at the most fundamental level of ontological description, serving as the primitive arbiter of temporal succession in the absence of any deeper or more elemental mechanism.

$\hat{H}\Psi = 0$ does not embody any intrinsic error in its formulation; rather, it stands as radically incomplete with respect to the full architecture of temporal dynamics. This equation accurately encodes the constraint that every valid state U_{t_L} must satisfy, namely that $\hat{H}$ annihilates the wavefunction associated with that state, thereby enforcing the diffeomorphism invariance and constraint algebra inherent to background-independent theories. However, the equation remains entirely silent regarding the dynamical origin of the sequence itself, offering no mechanism to generate the progression from one constrained state to the next. The Global Sequencer rectifies this deficiency by supplying the missing dynamical rule: $\mathcal{U}$ acts to map any Wheeler-DeWitt-constrained state to another state that likewise satisfies the Wheeler-DeWitt constraint, ensuring that the constraint propagates invariantly across the entire sequence. As a direct consequence, the total wavefunction of the universe cannot be construed as a single, timeless entity Ψ devoid of internal structure; instead, it manifests as an ordered history $\{\Psi[U_{t_L}]\}_{t_L=0}^{\infty}$, wherein the constraint $\hat{H}\Psi[U_{t_L}] = 0$ holds locally within logical time at every discrete step t_L , thereby reconciling the static constraint with the dynamical reality of succession.

1.2.2.1 Commentary: Ontological Status

Classification of the Sequencer Parameter as a Meta-Theoretical Operator

t_L does not qualify as a physical observable, in the sense that no measurement protocol within the physical system can yield its value; no coordinate embedded within the spacetime manifold; no field propagating through the configuration space; no degree of freedom that varies independently within the dynamical variables of the theory; and no integral part of the substrate from which states are constructed. t_L does not parametrize change within any state, instead, t_L exists as a purely formal, meta-theoretical iteration counter,

operating at a level of description that oversees and enumerates the computational steps without participating in their content or evolution. Its role parallels precisely the step number n in a Conway’s Game of Life simulation, where n merely indexes the generations of cellular updates without influencing the rules or states; or the renormalization scale μ in a holographic renormalization group flow, where μ parametrizes the coarse-graining hierarchy externally to the field theory itself; or the fictitious time τ employed in the Parisi-Wu stochastic quantization procedure, where τ drives the imaginary-time evolution as a non-physical auxiliary parameter; or the ontological time invoked in ’t Hooft’s Cellular Automaton Interpretation of quantum mechanics, where it discretely advances the hidden-variable substrate; or the unimodular time $\mathcal{T}$ introduced in the Henneaux-Teitelboim formulation of gravity, where $\mathcal{T}$ provides a global foliation parameter decoupled from local metrics. In each of these diverse frameworks (regardless of whether their respective authors have explicitly acknowledged the implication), an external, non-dynamical parameter covertly assumes the responsibility of generating succession, underscoring the ubiquity of such meta-temporal structures in foundational physical modeling.

1.2.2.2 Commentary: Computational Cosmology

Algorithmic Origins of Physical Law derived from Computational Universes

The operational nature of the Global Sequencer attains its most concrete and mechanistically detailed realization within the domain of discrete computational physics, particularly through the frameworks established by the Wolfram Physics Project (Wolfram, 2002); (Wolfram, 2020) and Gerard ’t Hooft’s Cellular Automaton Interpretation (CAI) of Quantum Mechanics. These frameworks furnish the essential conceptual and mathematical machinery required to effect a profound transition in the conceptualization of time: from a passive geometric coordinate subordinated to the metric tensor, to an active algorithmic process that orchestrates the discrete unfolding of relational structures.

Within the Wolfram model, the instantaneous state of the universe deviates fundamentally from the paradigm of a continuous differentiable manifold; instead, it materializes as a spatial hypergraph (a vast, dynamically evolving network comprising abstract relations among a multitude of nodes, where edges encode the primitive causal or adjacency connections). In this representational scheme, the “laws of physics” transcend the rigidity of static partial differential equations imposed on continuous fields; they instead embody a set of dynamic Rewriting Rules, which prescribe transformations on local substructures of the hypergraph. The evolution of the universe proceeds precisely as the algorithmic process of exhaustively scanning the hypergraph for occurrences of predefined target sub-patterns (for instance, a pairwise relation denoted as $\{A, B\}$ conjoined with $\{B, C\}$) and systematically replacing each such occurrence with a prescribed updated pattern, such as $\{A, C\}$ augmented by $\{A, B\}$. This rewriting operation, when applied in parallel across all eligible sites, generates the progression of states.

In this context, the Global Sequencer discharges the function of the Updater, coordinating the synchronous execution of all applicable rewrites within a given iteration. Each complete cycle of pattern identification and substitution delineates an “Elementary Interval” of logical time, during which the hypergraph undergoes a unitary transformation under the collective rule set. Time, therefore, does not “flow” as a continuous fluid medium susceptible to infinitesimal variations; rather, it “ticks” forward through a series of discrete updating events, each demarcated by the completion of the rewrite phase. The cumulative history of these successive updates coalesces into the Causal Graph, a directed acyclic structure that traces the precedence relations among elementary events; from this graph, the familiar macroscopic structures of relativistic spacetime (such as Lorentzian metrics, light cones, and geodesic paths) eventually emerge as effective approximations in the thermodynamic limit of large node counts. The Sequencer itself operates analogously to the “CPU clock” in a computational architecture, imposing a rhythmic discipline on the rewrite process and thereby converting the latent potential encoded within the initial rule set into the manifest actuality of an unfolding state history, replete with emergent complexity and observable phenomena.

In a parallel vein, ’t Hooft advances the position that the apparent indeterminism permeating standard formulations of Quantum Mechanics arises not as an intrinsic feature of nature but as an epistemic artifact stemming from the misapplication of continuous probabilistic superpositions to what is fundamentally a deterministic, discrete underlying mechanism. He delineates a sharp ontological distinction between the

“Ontic State” (a precise, unambiguous configuration of binary bits (or analogous discrete elements) realized at each integer value of time t , constituting the bedrock reality inaccessible to direct measurement) and the “Quantum State,” which serves merely as a statistical ensemble averaged over epistemic uncertainties, employed by observers whose instruments fail to resolve the granular updates of the ontic layer. Within this interpretive scheme, the universal evolution manifests as the action of a Permutation Operator $\hat{P}$, defined on the space of all possible ontic configurations and mapping this space onto itself in a bijective manner: $|\psi(t+1)\rangle = \hat{P}|\psi(t)\rangle$. This operator, by virtue of its discrete and exhaustive permutation of states, enacts precisely the role of the Global Sequencer: it constitutes the inexorable “cogwheel” mechanism that propels reality from one definite, ontically resolved configuration to the immediately succeeding one, thereby obviating any prospect of “timeless” stagnation or eternal superposition. The permutation ensures that succession occurs with absolute determinacy, aligning the discrete ticks of logical time with the emergence of quantum probabilities as mere shadows cast by incomplete observational access.

1.2.2.3 Commentary: Unimodular Gravity

Restoration of Unitarity by the Dynamical Cosmological Constant

Although computational models delineate the precise mechanism underlying the Global Sequencer, the physical justification for separating the Sequencer parameter (t_L) from the emergent geometric time (t_{phys}) draws robust and formal support from the theory of **Unimodular Gravity (UMG)**, with particular emphasis on the canonical quantization framework developed by Henneaux and Teitelboim. This theoretical edifice extracts the concept of a global time parameter from the paralyzing “frozen formalism” endemic to standard General Relativity, wherein the diffeomorphism constraints render time evolution illusory.

In the canonical formulation of standard General Relativity, the cosmological constant Λ enters the action as an immutable, fixed parameter woven into the fabric of the Einstein field equations, dictating the global curvature scale without dynamical variability. Unimodular Gravity fundamentally alters this paradigm by promoting Λ to the status of a dynamical variable (more precisely, by interpreting it as the canonical momentum conjugate to an independent spacetime volume variable, often denoted as the total integrated 4-volume). This promotion establishes a canonical conjugate pair, $[\hat{\Lambda}, \hat{\mathcal{T}}] = i\hbar$, wherein the commutator encodes the quantum uncertainty inherent to non-commuting observables. Here, the Unimodular Time variable $\mathcal{T}$ assumes the role of the “position-like” coordinate, while Λ functions as its “momentum-like” counterpart; given that Λ governs the vacuum energy density permeating empty spacetime, its conjugate $\mathcal{T}$ correspondingly tracks the cumulative accumulation of 4-volume across the cosmological expanse, thereby furnishing a global, objective metric for the universe’s elapsed “run-time” that transcends local gauge choices.

This canonical structure achieves the restoration of unitarity to the formalism of quantum cosmology, which otherwise succumbs to the atemporal constraints of general covariance. In the conventional approach to quantum gravity, $\hat{H}$ imposes a primary constraint demanding $\hat{H}\Psi = 0$ on the physical state space, thereby projecting the dynamics onto a subspace where time evolution vanishes identically and yielding the infamous frozen ‘Block Universe,’ in which all configurations coexist in a static, changeless totality devoid of intrinsic becoming (Rovelli & Smolin, 1990). By contrast, the incorporation of the dynamical time variable $\mathcal{T}$ within Unimodular Gravity perturbs the underlying constraint algebra, elevating the temporal progression to a first-class dynamical principle. The resultant equation of motion assumes the canonical form of a genuine Schrodinger equation parametrized by $\mathcal{T}$:

$$i\hbar \frac{\partial \Psi}{\partial \mathcal{T}} = \hat{H} \Psi$$

This evolution equation governs a state vector $|\Psi(\mathcal{T})\rangle$ that advances unitarily with respect to the affine parameter $\mathcal{T}$, preserving probabilities and inner products across increments in $\mathcal{T}$ while permitting the coherent accumulation of phases and amplitudes. The parameter $\mathcal{T}$ thereby incarnates the physical referent of the Global Sequencer within the gravitational sector: it operates in a “de-parameterized” mode, signifying its independence from the arbitrary local coordinate systems (or gauges) adopted by internal observers, who perceive only the relational t_{phys} derived from light signals and rod-and-clock measurements.

This separation of temporal scales aligns seamlessly with the principles of Lee Smolin’s Temporal Naturalism, which systematically critiques the Block Universe ontology (characterized by the eternal, simultaneous existence of past, present, and future) as profoundly incompatible with the empirical reality of quantum evolution, wherein unitary transformations manifest genuine change and contingency. Smolin contends that time must occupy a fundamental ontological status, irreducible to an emergent illusion, and that the laws of physics themselves may undergo evolution across cosmological epochs, thereby demanding a dynamical framework capable of accommodating such variability. The Global Sequencer (t_L), when physically instantiated as the Unimodular Time ($\mathcal{T}$), delivers precisely this preferred foliation: it enforces a universal slicing of the state sequence that underwrites the reality of the present moment, while preserving the local Lorentz invariance experienced by inertial observers, who remain ensconced within their parochial geometric clocks and precluded from discerning the meta-temporal progression.

1.2.2.4 Commentary: Background Independence

Independence of the Sequencer from Emergent Geometric Foliations through Pre-Geometric Definitions

Precisely because t_L resides at an external and non-dynamical stratum of the theory (untouched by the variational principles or symmetries governing the physical content), the entirety of the theory’s physical articulation (encompassing the relational linkages, correlation functions, and entanglement architectures intrinsic to each individual state U_{t_L}) remains utterly independent of any preferred time slicing, foliation scheme, or presupposed background manifold structure. All observables within the theory, ranging from scalar invariants to tensorial quantities like the emergent metric tensor and its associated Riemann curvature, derive their definitions and values exclusively from the internal relational properties and covariance relations obtaining within each U_{t_L} , without recourse to extrinsic coordinates or auxiliary geometries.

The Sequencer thus qualifies as pre-geometric in its essence: it inaugurates the genesis of geometric structures through the iterative application of relational updates, rather than presupposing their prior existence as a scaffold for dynamics, thereby upholding the stringent demands of manifest background independence characteristic of quantum gravity theories. Because t_L is a purely algebraic sequencing index devoid of metric structure, topological dimension, or coordinate geometry, the physical spacetime manifold—including Lorentzian intervals and proper time—emerges relationally from the causal poset of discrete events ($\mathcal{E}$), ensuring that no pre-existing geometric background is assumed.

1.2.2.5 Commentary: Page-Wootters Comparison

Superiority of the Sequencer Mechanism due to the Elimination of Clock Decoherence

The canonical Page-Wootters mechanism, which posits the total wavefunction of the universe as an entangled superposition of clock and system degrees of freedom wherein subsystem evolution emerges conditionally from the global constraint, harbors three fatal defects that undermine its foundational viability as a complete resolution to the problem of time:

1. **Ideal-clock assumption:** In realistic physical implementations, any candidate clock subsystem inevitably undergoes decoherence due to environmental interactions, thereby entangling with the observed system and inducing non-unitary evolution that dissipates coherence and violates the preservation of inner products and probabilities required for faithful timekeeping.
2. **Multiple-choice problem:** The partitioning of the total Hilbert space into a “clock” subsystem and a “system” subsystem admits a proliferation of inequivalent choices, each yielding distinct conditional evolution operators; these operators fail to commute or align, generating observer-dependent descriptions that lack universality and invite inconsistencies across different experimental contexts.
3. **Absence of genuine becoming:** The total state persists as an eternal, unchanging block configuration encompassing the entire history in superposition; what masquerades as “evolution” reduces to the computation of conditional probabilities within this preordained totality, precluding any ontological transition from potentiality to actuality and rendering change illusory.

t_L obviates all three defects in a unified stroke, restoring a robust ontology of temporal becoming:

- The operative “clock” resides at the meta-theoretical level and thus achieves perfection by constructive fiat, immune to decoherence, entanglement, or operational failure.
- Uniqueness inheres in the Sequencer by design; no multiplicity of alternatives exists, as it constitutes the singular, canonical iterator governing the universal state sequence.
- The update process effected by the Sequencer qualifies as an objective physical transition, wherein uncomputed potential configurations crystallize into definite, actualized states through the deterministic application of $\mathcal{U}$, thereby instantiating genuine novelty and diachronic identity.

Internal observers, operating within the emergent physical time t_{phys} , reconstruct the Page-Wootters conditional probabilities as an effective, approximate description valid in the regime of weak entanglement and coarse-grained measurements; however, the foundational ontology embeds authentic evolution, wherein each tick of t_L marks an irrevocable advance from one ontically distinct reality to the next (Page & Wootters, 1983); (Gambini, Garcia-Pintos, & Pullin, 2023).

1.2.3 Lemma: Finite Information Substrate

Finiteness and Quadratic Boundedness of the Information Substrate

Let t_L denote a finite logical time. Then the information content $S(U_{t_L})$ is strictly finite, and the growth of this content is bounded by a quadratic function of logical time, $S(U_{t_L}) \leq \mathcal{O}(t_L^2)$.

1.2.3.1 Proof: Finite Information Substrate

Derivation of the Quadratic Entropy Bound via Inductive Branching

I. Setup and Assumptions

Let Ω_t denote the set of admissible physical states at logical time t . Let $S(U_t) = \log_2 |\Omega_t|$ quantify the information content.

The physical postulates impose the following growth constraints:

1. **Finite Local Branching (b):** The **Finite Nature Hypothesis** limits the update capacity of the substrate. The number of physically distinct successor states for any state U is bounded by the local branching factor b raised to the number of active sites.

$$\forall U \in \Omega, \quad |\{U' \mid U \xrightarrow{\mathcal{U}} U'\}| \leq b^{s_t}$$

2. **Holographic Surface Scaling (δ):** The **Bousso Bound** restricts the number of active degrees of freedom to the surface area of the causal graph. This area s_t scales linearly with the radius in a discrete graph growing from a root.

$$s_t \leq \delta \cdot t \quad \text{where } \delta > 0$$

II. Derivation

The cardinality of the state space at step $t + 1$ is bounded by the product of the previous cardinality and the successor count defined by the branching factor and active sites.

$$|\Omega_{t+1}| \leq |\Omega_t| \cdot b^{s_t}$$

Logarithmic transformation converts this product into a summation for entropy calculation:

$$\log_2 |\Omega_{t+1}| \leq \log_2 |\Omega_t| + \log_2 (b^{s_t})$$

$$S(U_{t+1}) \leq S(U_t) + s_t \log_2 b$$

Let $\Delta S_t = S(U_{t+1}) - S(U_t)$. Substitution of the **Holographic Surface Scaling** constraint yields the explicit bound:

$$\Delta S_t \leq (\delta t) \log_2 b$$

III. Accumulation

The total entropy at time T is the sum of the initial entropy and all incremental changes.

$$S(U_T) = S(U_0) + \sum_{t=0}^{T-1} \Delta S_t$$

The unique primordial vacuum at $t = 0$ establishes the **Base Case**:

$$|\Omega_0| = 1 \implies S(U_0) = 0$$

Substitution of the derived bound for ΔS_t into the cumulative sum produces:

$$S(U_T) \leq 0 + \sum_{t=0}^{T-1} (\delta t \log_2 b)$$

Factoring out the time-independent constants $C = \delta \log_2 b$ isolates the arithmetic series:

$$S(U_T) \leq C \sum_{t=0}^{T-1} t$$

IV. Resolution and Conclusion

The arithmetic series evaluates via the standard summation formula with $n = T - 1$:

$$\sum_{t=0}^{T-1} t = \frac{(T-1)((T-1)+1)}{2}$$

$$\sum_{t=0}^{T-1} t = \frac{(T-1)T}{2}$$

$$\sum_{t=0}^{T-1} t = \frac{T^2 - T}{2}$$

Substitution of this result back into the entropy inequality yields:

$$S(U_T) \leq C \cdot \left(\frac{T^2 - T}{2} \right)$$

$$S(U_T) \leq \frac{\delta \log_2 b}{2}(T^2 - T)$$

For $T > 1$, the quadratic term strictly dominates the linear term, such that $T^2 - T < T^2$. This dominance relation establishes the upper bound:

$$S(U_T) < \frac{\delta \log_2 b}{2}T^2$$

We conclude that the information content growth is bounded by a quadratic function of logical time:

$$S(U_{t_L}) \leq \mathcal{O}(t_L^2)$$

This scaling holds for any locally finite, causally expanding graph.

Q.E.D.

1.2.4 Lemma: Backward Accumulation

Exclusion of Unbounded Past Direction

Assume the domain of the global logical time parameter T extends to the infinite past. Then this unbounded configuration is excluded by the **Finite Information Substrate**.

1.2.4.1 Proof: Backward Accumulation

Derivation of Contradiction via Entropy and Capacity Divergence

I. Setup and Assumptions

Let the temporal domain be unbounded in the past direction, denoted $T = \mathbb{Z}_{\leq 0}$. Let the history of the universe be the infinite sequence of states $\mathcal{H} = \{\dots, U_{-n}, \dots, U_{-1}, U_0\}$.

II. Case A: Irreversible Dynamics

Let $\mathcal{U}$ be a dissipative operator satisfying the Second Law of Thermodynamics. Let $\Delta S_k = S(U_{k+1}) - S(U_k)$ denote the entropy production at step k .

1. **Thermodynamic Positivity:** For non-equilibrium evolution involving coarse-graining or erasure, the expected entropy production is strictly positive:

$$\mathbb{E}[\Delta S_k] = \mu > 0$$

The fluctuations are bounded by the **Finite Information Substrate**:

$$\text{Var}(\Delta S_k) = \sigma^2 < \infty$$

2. **Cumulative Summation:** The total entropy at the present $t = 0$ is the accumulation of all prior productions. Let S_n denote the sum over the past n steps:

$$S_n = \sum_{k=-n}^{-1} \Delta S_k$$

3. **Probabilistic Divergence:** We apply Chebyshev's Inequality to bound the deviation of the time-averaged entropy production from the mean μ :

$$\mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \epsilon \right) \leq \frac{\sigma^2}{n\epsilon^2}$$

The limit $n \rightarrow \infty$ drives the probability of deviation to zero:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \epsilon \right) = 0$$

This implies almost sure convergence of the sum to the linear growth trend:

$$S(U_0) \approx \lim_{n \rightarrow \infty} n\mu = \infty$$

4. **Contradiction:** The divergence $S(U_0) \rightarrow \infty$ is excluded by the **Finite Information Substrate**.

III. Case B: Reversible Dynamics

Let $\mathcal{U}$ be a strictly unitary (bijective) operator.

$$U_{t+1} = \mathcal{U}(U_t) \iff U_t = \mathcal{U}^{-1}(U_{t+1})$$

1. **Injectivity of History:** The requirement of a non-cyclic history implies injectivity of the mapping from time to state:

$$\forall t_a, t_b \in T, \quad t_a \neq t_b \implies U_{t_a} \neq U_{t_b}$$

2. **Information Preservation:** In a deterministic reversible system, unitarity requires that the present state U_0 encode the unique trajectory of the past. Let ΔI_k denote the unique information bit distinguishing state U_{-k} from any other state in the sequence:

$$\Delta I_k \geq 1 \text{ bit}$$

3. **Capacity Aggregation:** The total information capacity required for U_0 to distinguish an infinite set of unique predecessors is the sum of these contributions:

$$I(U_0) \geq \sum_{k=1}^{\infty} \Delta I_{-k}$$

We evaluate the sum:

$$I(U_0) \geq \sum_{k=1}^{\infty} 1 = \infty$$

4. **Contradiction:** An infinite information capacity $I(U_0) = \infty$ is excluded by the **Finite Information Substrate**.

IV. Conclusion

Both dynamical regimes necessitate an infinite information content in the present state U_0 given an infinite past. We conclude that the temporal domain is bounded by a finite origin.

Q.E.D.

1.2.5 Lemma: Finite State Recurrence

Incompatibility of Infinite Past Duration with Strictly Finite Configuration Spaces

Assume the configuration space Ω possesses strictly finite cardinality. Then an infinite past trajectory necessitates a state recurrence that forms a closed causal loop, violating **Acyclic Effective Causality**.

1.2.5.1 Proof: Finite State Recurrence

Demonstration of Inevitable Causal Loops via the Dirichlet Principle

I. Setup and Assumptions

Let Ω denote the universal configuration space of admissible states. Assume the cardinality of this state space is strictly finite:

$$|\Omega| = N < \infty$$

II. The Infinite Past Hypothesis

Assume the temporal domain extends to the infinite past. Let the history of the universe correspond to a sequence of states indexed by non-positive logical time:

$$\mathcal{T} = (\dots, U_{-2}, U_{-1}, U_0)$$

Consider a finite subsequence of this history with length $N + 1$:

$$\mathcal{T}_{sub} = (U_{-N}, \dots, U_0)$$

Let $T = \{-N, \dots, 0\}$ denote the set of time indices for this subsequence, such that $|T| = N + 1$.

III. Application of the Dirichlet Principle

Let $f : T \rightarrow \Omega$ define the mapping $f(t) = U_t$. Comparison of the domain cardinality $|T| = N + 1$ and the codomain cardinality $|\Omega| = N$ reveals that the mapping cannot be injective. The Dirichlet (Pigeonhole) Principle implies the existence of at least two distinct time indices $t_a, t_b \in T$ with $t_a < t_b$ such that the system occupies identical states:

$$U_{t_a} = U_{t_b}$$

IV. Deterministic Evolution and Cycle Formation

Let $\mathcal{U}$ denote the deterministic evolution operator satisfying $U_{t+1} = \mathcal{U}(U_t)$. The identity of the states U_{t_a} and U_{t_b} implies the identity of their successors:

$$\mathcal{U}(U_{t_a}) = \mathcal{U}(U_{t_b}) \implies U_{t_a+1} = U_{t_b+1}$$

Mathematical induction extends this identity to all subsequent steps $k \geq 0$, establishing $U_{t_a+k} = U_{t_b+k}$. The trajectory enters a periodic cycle of length $P = t_b - t_a$:

$$C = (U_{t_a}, U_{t_a+1}, \dots, U_{t_b-1})$$

This recurrence establishes the following closed causal structure:

$$U_{t_b-1} \rightarrow U_{t_b} \equiv U_{t_a}$$

$$U_{t_a} \rightarrow U_{t_a+1} \rightarrow \dots \rightarrow U_{t_b-1} \rightarrow U_{t_a}$$

V. Contradiction with Acyclicity

The existence of the cycle C implies that the state U_{t_a} constitutes a causal ancestor of itself ($U_{t_a} \prec U_{t_a}$). This transitive self-reference violates **Acyclic Effective Causality**. We conclude that an infinite past acyclic trajectory is incompatible with a strictly finite configuration space.

Q.E.D.

1.2.6 Lemma: Supertask Impossibility

Impossibility of Infinite Operation Sequences from Logical and Physical Non-Termination

The traversal of an infinite sequence of discrete computational steps to arrive at the present state U_0 constitutes a Supertask. The completion of a Supertask is physically undefined within the dynamical constraints of the theory, as it requires the execution of $\aleph_0$ operations in finite time or the existence of a completed infinity. Neither is permissible in a constructive ontology.

1.2.6.1 Proof: Supertask Limits

Formal Proof of Non-Termination via Turing Computability and Relativistic Constraints

I. Definition of the History Sequence

Let the history $\mathcal{H}$ be defined as the ordered set of computational operations $\mathcal{U}_i$ required to generate the present state U_0 from a precedent state. Under the hypothesis of an infinite past ($t \in \mathbb{Z}_{\leq 0}$), the index set is the negative integers $\mathbb{Z}_{\leq -1}$.

$$\mathcal{H} = \{\dots, \mathcal{U}_{-3}, \mathcal{U}_{-2}, \mathcal{U}_{-1}\}$$

This set possesses the order type ω^* (the order of the negative integers), which is characterized by having a last element ($\mathcal{U}_{-1}$) but no first element.

II. The Supertask Constraint

For the state U_0 to be physically realized (to exist as the output of a computation), the entire sequence of operations in $\mathcal{H}$ must have been executed to completion. This implies the performance of a **Supertask**: an infinite number of discrete steps completed within the timeline prior to $t = 0$.

III. Computational Undefinability (The Initialization Problem)

We model the physical universe as a State Machine $M = (S, \Sigma, \delta, s_0)$, where s_0 is the initial state.

1. **Requirement:** For any computation to proceed, the machine must be initialized in state s_0 at some time t_{start} .
2. **Deficiency:** In the sequence $\mathcal{H}$, for any hypothesized starting time t_k , there exists a prior operation $\mathcal{U}_{t_k-1}$ that was required to generate the input for $\mathcal{U}_{t_k}$.

$$\forall k \in \mathbb{Z}, \quad \exists (k-1) \in \mathbb{Z} \quad \text{such that} \quad k-1 < k$$

3. **Result:** There is no time t at which the machine M could have been initialized.

$$\text{Domain}(\mathcal{H}) \cap \{t_{start}\} = \emptyset$$

A computation with no initial state is mathematically undefined.

IV. Energy Divergence (The Resource Problem)

Let $\epsilon(op)$ be the energy cost of a single logical operation. By Landauer’s Principle and the Margolus-Levitin theorem, any state transition takes a non-zero amount of energy and time.

$$\epsilon(op) \geq \epsilon_{min} > 0$$

The total energy E_{total} dissipated to reach state U_0 is the sum over the infinite history:

$$E_{total} = \sum_{k \in \mathcal{H}} \epsilon(\mathcal{U}_k)$$

Since the sequence is infinite and the terms are bounded below by ϵ_{min} :

$$E_{total} \geq \sum_{k=1}^{\infty} \epsilon_{min} = \lim_{n \rightarrow \infty} (n \cdot \epsilon_{min}) = \infty$$

An infinite energy dissipation implies that the universe must have exhausted all free energy (reached thermodynamic equilibrium) infinitely long ago. This contradicts the existence of the low-entropy, ordered state U_0 observed at the present.

Q.E.D.

1.2.6.2 Commentary: Collapse of Supertasks

Dynamical Instability of Infinite Computation due to General Relativistic Constraints

The logical impossibility inherent to an infinite past finds a precise physical counterpart in the phenomenon designated as the **Gravitational Collapse of Supertasks**, a dynamical instability wherein the machinery postulated to execute such a transfinite computation self-destructs under general relativistic backreaction. As demonstrated by Gustavo Romero in 2014, the apparatus required to perform an infinite sequence of operations (thereby “arriving” at the present from an eternal regress) inevitably succumbs to singularity formation prior to completion.

This collapse arises from the interplay of two inexorable physical limits, each amplifying the other’s effects to catastrophic divergence:

1. **Landauer’s Principle:** Every irreversible logical operation, such as bit erasure or conditional branching in the Sequencer’s update rules, incurs a minimal thermodynamic cost of $E \geq k_B T \ln 2$ in dissipated heat (Landauer, 1991); (Bennett, 1982), where T denotes the ambient temperature of the computational substrate. For an infinite sequence of steps, assuming a constant (or even diminishing) energy per operation $\epsilon > 0$, the cumulative energy expenditure integrates to $E_{total} = \sum_{k=-\infty}^0 \epsilon_k \rightarrow \infty$, demanding an unbounded reservoir that no finite universe can supply without violating the first law of thermodynamics. This thermodynamic limit maps directly to the pre-geometric substrate: “energy” corresponds structurally to the algebraic operation count (computational cost) required to modify the relational network, while “temperature” represents the dimensionless scaling parameter of the graph’s partition function. A logically irreversible edge deletion () thus redistributes structural degrees of freedom, generating local entropic topological noise (the discrete analogue of heat) that would, in an infinite regress, accumulate without bound and prevent the nucleation of a stable pre-geometric spatial structure.
2. **Heisenberg Uncertainty:** To confine the infinite sequence within a finite elapsed coordinate time (or to “reach” the present from an eternal regress), the temporal allocation per step must contract to $\Delta t_k \rightarrow 0$ as $k \rightarrow -\infty$. The time-energy uncertainty relation $\Delta E \Delta t \geq \hbar/2$ then mandates that energy fluctuations scale inversely: $\Delta E_k \geq \hbar/(2\Delta t_k) \rightarrow \infty$. These fluctuations, manifesting as virtual

particle-antiparticle pairs or vacuum polarization in quantum field theory, engender unbounded energy densities within the localized computing region.

Within the framework of **General Relativity**, localized energy concentrations serve as the gravitational source term in the Einstein field equations $G_{\mu\nu} = 8\pi GT_{\mu\nu}/c^4$; the accumulation of infinite total energy (or infinite density from quantum fluctuations) thus warps spacetime with ever-increasing curvature. The Schwarzschild radius $R_s = 2GM/c^2$, where M quantifies the enclosed mass-energy, swells without bound as $M \rightarrow \infty$. Inevitably, R_s surpasses the physical extent of the computational domain (say, the horizon of the observable universe or the causal patch of the Sequencer), triggering the formation of an event horizon. Beyond this threshold, the system implodes into a black hole singularity, where geodesics terminate and information retrieval becomes impossible.

This inexorable collapse precludes the universe from “computing” an infinite history to manifest the present, as the requisite machinery gravitationally annihilates itself mid-task, prior to outputting a coherent “Now.” The empirical persistence of a stable, non-singular present configuration (evidenced by the absence of horizon encirclement and the continuity of cosmic evolution) thus constitutes irrefutable proof that the past admits no infinite regress; the temporal domain must commence at a finite origin to evade such dynamical catastrophe.

1.2.7 Theorem: Temporal Finitude

Necessity of a Finite Temporal Origin demanded by the Logical Exclusion of Infinite Regress

The domain of Global Logical Time t_L is strictly lower-bounded. There exists a unique initial state, designated U_0 , which possesses no causal predecessor. The domain of t_L is isomorphic to the set of non-negative integers $\mathbb{N}_0$, establishing a definite moment of genesis for the computational process.

1.2.7.1 Proof: Temporal Finitude

Temporal Finitude

I. The Infinite Hypothesis Let it be assumed, for the explicit purpose of demonstrating a contradiction, that the domain of Global Logical Time t_L is unbounded in the past direction. This assumption implies that the set of temporal indices is isomorphic to the non-positive integers ($T_L \cong \mathbb{Z}_{\leq 0}$), thereby asserting the existence of an infinite sequence of distinct antecedent states $\{\dots, U_{-2}, U_{-1}, U_0\}$.

II. The Constraint Chain The validity of this hypothesis is interrogated against the established lemmas of the theory:

1. **Finite Information Substrate** : The system enforces a strict holographic bound on the information content of any state within the sequence. It is established that $S(U_t)$ must remain finite for all finite t . The assumption of an infinite past requires the current state to encode a history of infinite depth, which necessitates an information capacity that exceeds this finite bound.
2. **Backward Accumulation** : Under the condition of irreversible dynamics, an infinite past necessitates an unbounded accumulation of entropy production ($\Sigma \Delta S \rightarrow \infty$). This accumulation would result in a present state U_0 characterized by maximal entropy (Thermodynamic Equilibrium or Heat Death), a condition that stands in direct contradiction to the observed low-entropy configuration of the physical universe.
3. **Finite State Recurrence** : Under the condition of reversible dynamics within a state space of finite cardinality, an infinite temporal duration necessitates the occurrence of Poincare recurrence ($U_t = U_{t+k}$). Such recurrence establishes closed causal loops, which constitute a direct violation of the **Acyclicity** axiom governing the causal graph.
4. **Supertask Impossibility** : The logical traversal of an infinite sequence of operations to arrive at the present state U_0 constitutes a Supertask. The completion of such a task is computationally undefined, as it lacks a valid initialization condition, rendering the existence of U_0 logically impossible under constructive dynamical rules.

III. Convergence The assumption of an unbounded past generates inescapable contradictions under both thermodynamic and computational constraints. Whether the dynamics are reversible or irreversible, the hypothesis fails to yield a consistent physical model.

IV. Formal Conclusion Consequently, the temporal domain cannot be unbounded. There must exist a unique initial state U_0 such that for all integers $t < 0$, the state U_t is undefined. The domain of Global Logical Time is isomorphic to the set of non-negative integers $\mathbb{N}_0$, thereby establishing a definite and absolute moment of genesis.

Q.E.D.

1.2.7.2 Commentary: Grim Reaper Paradox

Logical Necessity of Finite Temporal Origins demonstrated by the Grim Reaper Paradox

The assertion that the Global Sequencer demands a definite starting point ($t_L = 0$), precluding any infinite regress, garners unassailable logical reinforcement from the **Grim Reaper Paradox** (originally formulated by Jose Benardete and subsequently fortified through the analytic refinements of Alexander Pruss and Robert Koons). This paradox furnishes a formal, a priori proof for **Causal Finitism**, the foundational axiom decreeing that the historical trajectory of any causal system cannot extend to an actual infinity in the backward direction, as such an extension vitiates the chain of sufficient reasons.

Envision a hypothetical universe inhabited by a single victim, designated Fred, alongside a countably infinite ensemble of Grim Reapers $\{R_1, R_2, R_3, \dots\}$, each programmed with an execution protocol contingent on Fred's survival. The drama unfolds within the temporal interval spanning 12:00 PM to 1:00 PM, with assignments calibrated to converge supertask-wise:

- Reaper R_1 activates at precisely 1:00 PM, tasked with killing Fred should he remain alive at that instant.
- Reaper R_2 activates at 12:30 PM (midway to 1:00 PM), similarly conditioned on Fred's survival to that earlier threshold.
- In general, Reaper R_n activates at the epoch $12 : 00 + (1/2)^{n-1}$ hours PM, executing the kill if Fred persists alive upon its arrival.

As the index n ascends to infinity, the activation epochs form a convergent geometric series: $t_n = 12 : 00 + \sum_{k=1}^{n-1} (1/2)^k$ hours, with $\lim_{n \rightarrow \infty} t_n = 12 : 00$ PM approached asymptotically from the future side. This setup prompts two innocuous interrogatives concerning Fred's status at 1:01 PM, each exposing the paradox's barbed core:

1. **Is Fred dead?** Affirmative. Survival beyond 1:00 PM proves impossible, as Reaper R_1 (the coarsest sentinel) guarantees termination at or before that boundary; no prior reaper can avert this, and the ensemble collectively overdetermines the outcome.
2. **Which Reaper killed him?** Indeterminate by exhaustive elimination. Suppose, per absurdum, that Reaper R_n effects the kill at t_n . This supposition entails Fred's aliveness immediately antecedent to t_n , permitting R_n 's conditional trigger. Yet Reaper R_{n+1} , stationed at $t_{n+1} = t_n - (1/2)^n$ hours (strictly prior), would have encountered that aliveness and preemptively executed, rendering R_n 's opportunity moot. This regress applies recursively: no finite n sustains the supposition, as each defers to a denser predecessor.

The resultant impasse manifests a closed causal loop: the terminal effect (Fred's death) stands guaranteed by the infinite assembly, yet its proximal cause (the executing reaper) eludes identification within the countable set, dissolving into logical vacuity. The death precipitates as a "brute fact" (an occurrence destitute of mechanistic ancestry, flouting the Principle of Sufficient Reason by which every contingent event traces to a determinate precursor). This configuration unveils the **Unsatisfiable Pair Diagnosis**: the conjoined propositions of an infinite past and causal consistency prove jointly untenable, as the former erodes the latter into paradox. Since the ontology of physics presupposes causal consistency (insisting that each state U_{t_L+1}

emerges as a well-defined function $f(U_{t_L}, \mathcal{U})$ of its antecedent and the evolution rule), we must excise the infinite past to preserve the chain's integrity. The Sequencer thus requires bounding below by a **First Event**, the uncaused cause (U_0) from which all subsequent effects descend with unambiguous pedigree, ensuring the historical manifold remains a tree-like arborescence rather than a gapped abyss.

The “Unsatisfiable Pair Diagnosis” (UPD), as articulated and defended by philosophers of time such as Alexander Pruss, reframes the perennial debate over temporal origins from speculative metaphysics to a logical dilemma. It diagnoses the paradoxes of infinite regress (exemplified by the Grim Reaper ensemble) not as idiosyncratic curiosities amenable to ad hoc dissolution, but as diagnostic indicators of a profound incompatibility between two axiomatic pillars that cannot coexist without mutual subversion.

1. The Logical Fork

The UPD compels a binary election between two elemental axioms, whose simultaneous affirmation generates inconsistency:

- **Axiom A (Infinite Past):** The temporal domain extends without lower bound, such that $t_L \in \mathbb{Z}_{\leq 0}$, admitting an actualized transfinite regress of prior states and events.
- **Axiom B (Causal Consistency):** The governance of physical events adheres to causal laws, encompassing local interaction Hamiltonians, the Markov property (future dependence solely on the present configuration), and the Principle of Sufficient Reason (every contingent occurrence admits a complete causal explication), thereby ensuring that effects inherit their necessity from identifiable antecedents.

2. The Conflict

Within the Grim Reaper tableau, endorsement of **Axiom A** (positing the actual existence of the infinite reaper sequence) precipitates the downfall of **Axiom B**. Fred's demise at or before 1:00 PM follows inexorably from the supertask convergence, yet the identity of the lethal agent proves logically inaccessible: it cannot devolve to Reaper R_1 (preempted by R_2), nor to Reaper R_2 (preempted by R_3), nor to any finite Reaper R_n (preempted by R_{n+1}), exhausting the possibilities without resolution.

This lacuna births a “**brute fact**” (the death eventuates sans specific causal agency, an *ex nihilo* irruption unmoored from the dynamical laws). Under infinite regress, causality fractures into “gaps,” wherein terminal effects manifest without proximal mechanisms, akin to spontaneous violations of unitarity or conservation. The infinite ensemble, while ensuring the outcome, dilutes responsibility across an uncompletable chain, rendering the causal narrative incomplete within the countable set.

3. The Priority of Physics

The discipline of physics dedicates itself to the elucidation of **Causal Consistency**, modeling phenomena through predictive functions that map initial data to outcomes via invariant laws. To countenance “uncaused effects” as a mere concession to the mathematical allure of an infinite past would eviscerate this enterprise: we could no longer assert that U_{next} derives deterministically (or probabilistically) from $U_{current}$, inviting arbitrariness and undermining empirical falsifiability. The scientific method, predicated on reproducible causation, demands the rejection of brute facts in favor of explanatory closure.

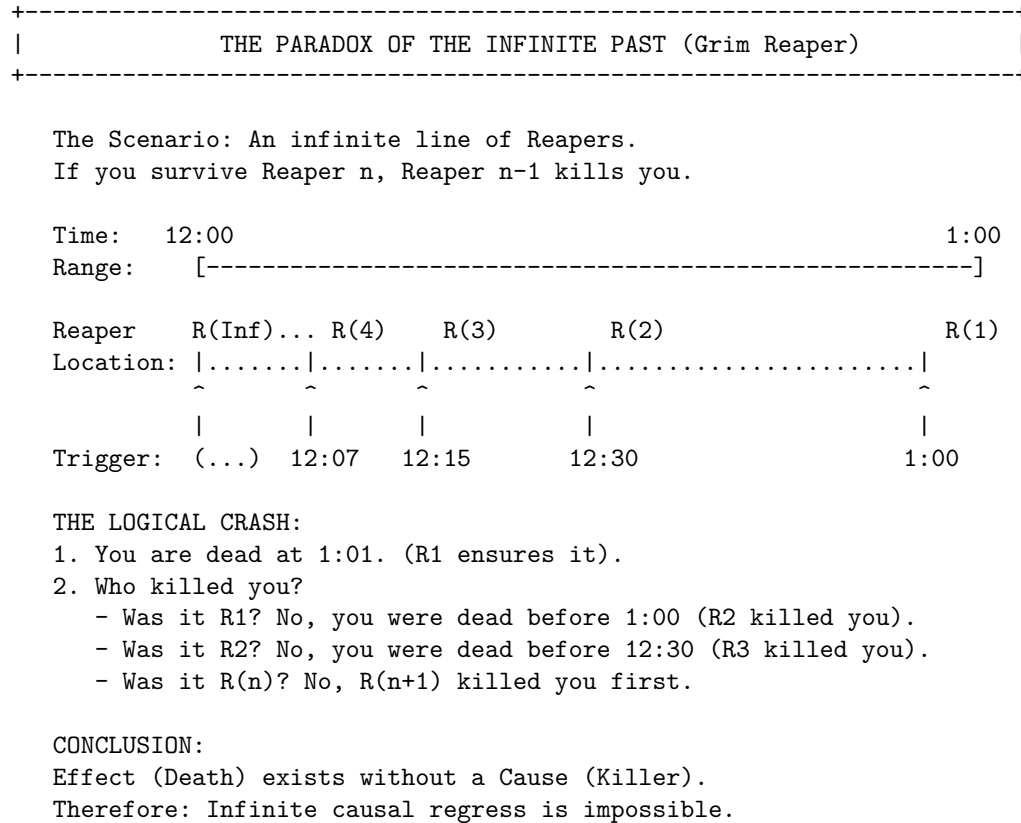
Conclusion

Empirical scrutiny confirms the universe's obeisance to causal laws (**Axiom B** enjoys verificatory status through the success of predictive theories from quantum electrodynamics to general relativity), while the UPD attests the mutual exclusivity of A and B. Ergo, **Axiom A** must yield to falsehood.

The universe thus mandates a **finite history**, with the Global Sequencer initiating at $t_L = 0$ to forge an unbroken causal spine: every event traces, through finite recursion, to the First Event U_0 , the axiomatic genesis beyond which no antecedents lurk. This finitistic resolution not only exorcises the Grim Reaper's specter but elevates the temporal ontology to a bastion of logical and physical coherence.

1.2.7.3 Diagram: Grim Reaper Paradox

Visualization of Asymptotic Convergence within the Grim Reaper Paradox



1.2.Z Implications and Synthesis

Temporal Ontology

Forcing the timeline to be finite cuts off the infinite regress. This ensures that every state possesses a definite causal ancestry traceable back to a singular origin. The conclusion is inescapable. “Becoming” is a discrete process. It is a sequence of state transitions that can be counted but not divided. This eliminates the possibility of a universe that has always existed. It grounds physics in a definite genesis where the first state acts as the uncaused cause of the computational chain. It implies that the history of the universe is a finite string of data. It is fully enumerable and logically bounded. This prevents the singularities associated with infinite pasts.

The logical clock t_L emerges here not as a coordinate dimension that one can travel through. It emerges as the relentless driver of existence itself. It acts as the fundamental CPU cycle of the universe. It is an external iterator that processes the state transition function. This distinction is vital because it separates the act of change from the measurement of change. Physical time is the variable that appears in relativity equations and is measured by atomic clocks. It is an emergent property of the relations inside the graph and is subject to dilation and curvature. Logical time is the absolute ordering of the computation. It is immune to these relativistic effects. By separating these two concepts, we resolve the Problem of Time in quantum gravity. The universe has a heartbeat, but it is not a clock hanging on the wall of spacetime.

With the clock established, the nature of the object that evolves must be defined. We have secured the timing and the mechanism of the update cycle. However, a heartbeat requires a body to animate. Time cannot exist in a vacuum because it requires a state to transition from and to. We turn now to the definition

of the spatial substrate. We must define the graph that serves as the memory of the system. We must define the canvas upon which this temporal iterator paints the history of the cosmos.



1.3 Causal Graph

With the manifold removed, only points and connections remain for analysis. Defining a point without a location forces a shift in perspective. It requires that an object must exist solely through its relations to others. If position is not intrinsic, then identity must be derived. We must construct a reality where location is defined entirely by connectivity. We must ask how a universe can exist before there is a physical space for it to exist in. We are effectively bootstrapping geometry from pure algebra. This requires us to abandon the comfortable intuition of spatial embedding and accept a world of pure abstraction.

Our structure must derive identity entirely from connectivity. We treat the graph as the raw material of spacetime. In this model, the edges themselves carry the burden of causality. Each link represents a transfer of influence. It is a discrete quantum of connection that binds two events together. This web of relations is not a map of the territory. It is the territory itself. It serves as the physical memory of the system. It encodes the past interactions that define the present state. Without this rigorous definition, we risk smuggling in spatial assumptions that undermine the background independence of the theory.

Analysis is restricted to a specific class of graphs to ensure physical viability. Loops where an event serves as its own ancestor are structurally unsound. Vague connections that fail to specify a direction of influence are equally problematic. We identify the necessary components as abstract events and causal links. We attach logical time to these edges to create a permanent record of creation. Constructing a structure rigid enough to preserve history yet flexible enough to permit evolution, we define the state space not as a collection of positions but as a collection of relations. This formalization allows us to treat the universe as a mathematical object that can be updated and computed.



1.3.1 Definition: State Space and Graph Structure

Structure of the Universal State Space as a Collection of Finite Acyclic Directed Graphs

Ω comprises the set of all kinematically admissible graph configurations that satisfy the constraints of finiteness and acyclicity. Each configuration in Ω encodes an essential “moment” in the universe’s history, represented by a single point $G \in \Omega$, which captures the complete relational and temporal structure at that instant without presupposing prior states or future evolutions. The finiteness constraint limits $|V| < \infty$ for every G , ensuring computational tractability and avoiding infinities that could undermine the discrete genesis principle, while acyclicity enforces the strict forward direction of causation, precluding loops that would imply retroactive influences or paradoxes.

$G = (V, E, H)$ constitutes the essential structural unit of Ω . This triplet encapsulates the essential components of relational existence, where each element contributes to the graph’s representational power: V provides the discrete event basis, E the primitive causal linkages, and H the immutable temporal ordering.

- V : $V = \{v_1, v_2, \dots, v_N\}$ forms a finite collection of vertices, each representing an elementary **Abstract Event**. These vertices serve as the raw “atoms” of existence, possessing no internal structure, spatial extent, geometric coordinates, or intrinsic properties beyond their index. The finiteness of $N = |V|$ arises from the constructive dynamics of the theory, where events emerge sequentially rather than pre-existing eternally, ensuring that the state space remains countable and free from unphysical infinities. Abstract events embody the minimal ontological primitives: they lack duration or magnitude, functioning solely as placeholders for relational intersections, which allows the theory to prioritize causality over substantival attributes.

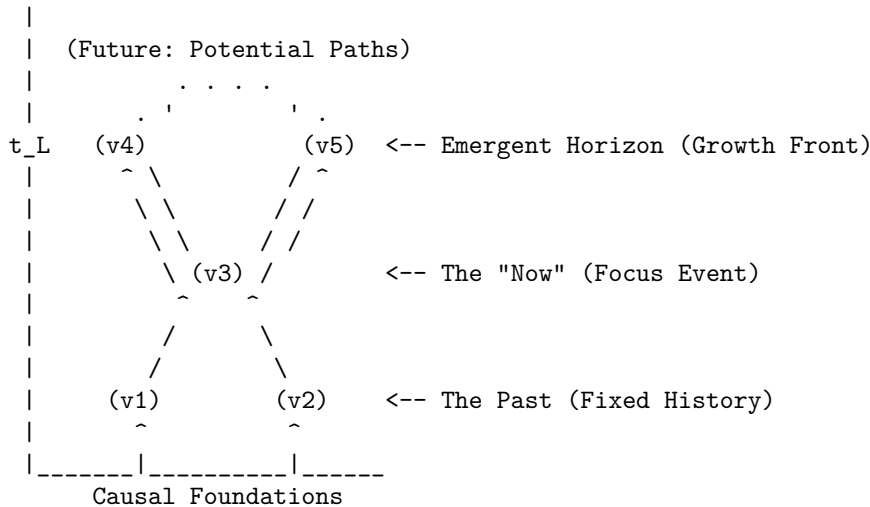
- $E: E \subseteq V \times V$ collects directed edges, each representing an irreducible **Causal Relation**. An edge $e = (u, v)$ asserts the primitive logical proposition “ u precedes v ,” denoting a direct, unmediated influence from event u to event v . Irreducibility means that no intermediate events intervene in the relation; if such mediation existed, the direct edge would decompose into a path of multiple edges, preserving the transitive closure under $\leq$ without loss of expressivity. The directed nature enforces asymmetry, aligning with the irreversible arrow of time, and the subset relation $E \subseteq V \times V$ permits sparsity (Bombelli et al., 1987); (Sorkin, 2005), reflecting the vacuum’s low density where most potential pairs remain unrealized until relational necessity demands them.
- $H: H : E \rightarrow \mathbb{N}$ assigns to each edge $e \in E$ a **Creation Timestamp**, drawn strictly from t_L at the instant of the edge’s formation during a dynamical tick. The codomain $\mathbb{N}$ (non-negative integers starting from 0) underscores the sequential, constructive nature of physical processes: timestamps increment monotonically ($H(e') > H(e)$ for edges formed later), recording the exact order of genesis without allowing continuous interpolation or retroactive assignment. This discreteness prevents paradoxes associated with infinite past histories or fractional times, as each edge receives its timestamp upon instantiation via **The Universal Constructor**, ensuring H embeds the full temporal archive immutably.

This triplet structure ensures that each $G \in \Omega$ represents a complete, self-contained snapshot of causal reality at a logical instant, with finiteness bounding complexity, acyclicity safeguarding consistency, and the history map providing an indelible record of emergence. The choice of $\mathbb{N}$ for H emphasizes the discrete genesis over continuous models, where time subdivides arbitrarily; here, the causal graph posits a punctuated history beginning from an initial empty state, avoiding logical paradoxes from pre-existing infinite chains and enabling dynamical evolution from nullity.

H is defined as an intrinsic attribute of the edge isomorphism class, not as a mutable data register. The timestamp is a topological invariant of the edge’s existence profile. Therefore, the “record” of an edge is not a separate resource that requires storage allocation; it is a fundamental definitional component of the edge itself. To delete an edge is to alter the graph topology, but the state space and graph structure of the deleted element remains mathematically distinct from a non-existent element due to its historical index.

1.3.1.1 Diagram: Causal Cone

Representation of Causal Horizons through the Emergent Growth Front



1.3.2 Definition: Emergent Timestamp Assignment

Assignment of Immutable Creation Timestamps by the Global Sequencer

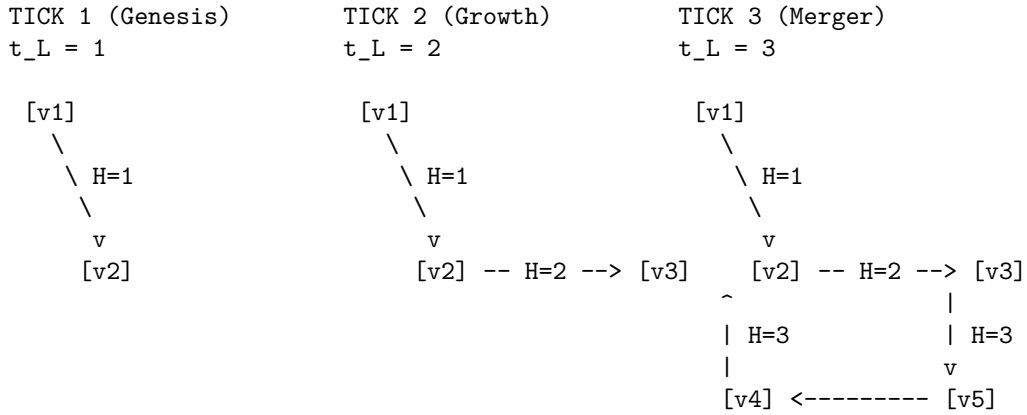
Time in Quantum Braid Dynamics operates as a persistent, immutable memory of creation embedded directly within the graph's structure. For any edge $e = (u, v)$ added to the graph during a dynamical tick at t_L , the **timestamp** $H(e)$ receives permanent assignment according to the current state of the Sequencer mechanism, defined in **Global Logical Time** :

$$H(e) = t_L$$

This assignment couples the ontology of the graph to the meta-theoretical Sequencer, which tracks the cumulative count of ticks since genesis. $H(e)$ constitutes an indelible record of origin: once the edge materializes via the rewrite rule, $H(e)$ fixes irrevocably, immune to subsequent modifications or retroactive adjustments. This immutability enables the full causal order to reconstruct solely from the graph's topological data, rendering the “flow” of time an intrinsic emergent property of the relations rather than an extrinsic parameter imposed upon the structure. The natural number codomain of H reinforces discreteness, with each increment marking a discrete genesis event, precluding continuous interpolation and ensuring the history forms a well-ordered sequence aligned with the theory's punctuated evolution.

1.3.1.2 Diagram: Timestamp Evolution

Illustration of Immutable Timestamp Assignment during Graph Evolution



RULE: $H(e_{\text{new}}) = t_L$ (Current Global Logical Time)

CONSTRAINT: $H(e)$ is immutable once assigned.

1.3.3 Definition: Abstract Event

Identity of the Abstract Event Vertex as a Purely Relational Nexus

An **Abstract Event** is a vertex $v \in V$. The identity of v is determined strictly by its relational connectivity within E . The vertex possesses no intrinsic properties, coordinates, or internal structure independent of these relations. It is a structureless point of intersection for causal influences.

1.3.3.1 Commentary: Relational Justification

Justification of Pre-Geometric Event Identity through Diffeomorphism Invariance

The relational ontology established by **Abstract Event** resolves the background dependence paradoxes inherent in classical physics by locating identity strictly within the links rather than the nodes. The abstract

event diverges fundamentally from a “point” in classical or Riemannian geometry. A geometric point derives identity from extrinsic coordinates embedded within a pre-existing background manifold, which serves as the substantive stage upon which dynamics unfold. In contrast, the abstract event in Quantum Braid Dynamics admits no such background. Its identity emerges purely relationally, defined exhaustively by the directed edges incident to it: outgoing edges designate it as cause, incoming as effect, with the degree sequence and timestamp offsets providing the sole descriptors.

For instance, in a minimal universe comprising two connected events $A \rightarrow B$, event A acquires no absolute position or intrinsic marker. Event A manifests relationally as “the direct cause of B ,” while event B manifests as “the direct effect of A .” The absence of self-attributes ensures that physics originates from the topology and dynamical evolution of the relations interconnecting them. This relational ontology aligns the foundational structure with the background-independent imperatives of quantum gravity theories, where spacetime arises as a derived construct from causal sets or spin networks rather than a primitive arena. The explicit exclusion of coordinates precludes substantivalism, enforcing diffeomorphism invariance at the discrete level: relabeling vertices preserves the causal skeleton, with isomorphism classes under edge-preserving maps defining equivalence. This shift from substantive objects to relational structures not only evades the hole argument but also embeds the theory’s discreteness, where events nucleate via edge additions, inheriting timestamps and influences solely from predecessors. This structure maintains a rigorous distinction between the event-level Causal History Graph—a strict Directed Acyclic Graph (DAG) by **Causal Ordering**—and the instantaneous Spatial State Graph G_t , which is tiled with directed 3-cycles representing geometric area. Because these spatial 3-cycles do not represent chronological loops in the event poset, spatial geometric triangles form without violating global causal acyclicity ($\emptyset$).

1.3.4 Theorem: Monotonicity of History

Strict Monotonicity of Causal Timestamp Sequences enforced by Recursive Assignment

The assignment of timestamps ensures that H induces a well-founded partial order on E . Specifically, for any newly created edge $e = (u, v)$, the timestamp satisfies the local recurrence relation:

$$H(e) = 1 + \max(\{H(e') \mid e' = (w, u) \in E\} \cup \{0\})$$

where the maximum ranges over all edges e' incoming to the source vertex u . If u admits no incoming edges (i.e., the set is empty, as occurs for isolated vertices in the initial vacuum state), the convention $\max(\emptyset) = 0$ applies, guaranteeing that primordial edges receive $H(e) = 1$. This recurrence enforces strict monotonicity of causality: no effect precedes its cause in the timestamp ordering, preserving the forward arrow of logical time across all transformations (Lamport, 1978).

1.3.4.1 Proof: Monotonicity

Formal Proof of Order Preservation from Inductive Stability

I. The Timestamp Assignment Algorithm

Let $\mathcal{C}$ be the constructor function responsible for edge creation. For any new edge $e = (u, v)$, the constructor assigns a timestamp $H(e)$ based on the strict causal history of the source vertex u . We define the set of incoming edges to u as $\text{In}(u) = \{e' \in E \mid e' = (w, u)\}$. The assignment rule is defined recursively:

$$H(e) = 1 + \max(\{H(e') \mid e' \in \text{In}(u)\} \cup \{0\})$$

II. The Irreflexivity Condition (Proof by Stability Analysis)

We test the stability of the timestamp assignment for a hypothetical self-loop edge $e_{self} = (u, u)$.

1. **Pre-computation:** The constructor queries the current history of u . Let the maximum existing timestamp be T_{max} .

$$T_{max} = \max(\{H(e') \mid e' \in \text{In}(u)_{\text{pre}}\} \cup \{0\})$$

The calculated timestamp for the new edge is:

$$H(e_{self}) = T_{max} + 1$$

2. **State Update:** If the edge e_{self} is added to the graph, the set of incoming edges updates:

$$\text{In}(u)_{\text{post}} = \text{In}(u)_{\text{pre}} \cup \{e_{self}\}$$

3. **Stability Constraint:** For the assignment to be valid, the rule must hold for the edge *after* it is added to the set.

$$H(e_{self}) > \max_{k \in \text{In}(u)_{\text{post}}} H(k)$$

4. **Substitution:** The maximum of the new set includes the edge itself.

$$\max_{k \in \text{In}(u)_{\text{post}}} H(k) = \max(T_{max}, H(e_{self}))$$

Since $H(e_{self}) = T_{max} + 1$, the maximum is $H(e_{self})$. Substituting back into the stability constraint:

$$H(e_{self}) > H(e_{self})$$

5. **Contradiction:** The inequality $x > x$ is false for all real numbers. Thus, no stable timestamp can be assigned to a self-loop. The operation creates a logical contradiction and is rejected by the constructor.

III. Transitive Order Preservation (Inductive Step)

We prove that for any causal path $\pi = (v_0, v_1, \dots, v_k)$, the sequence of edge timestamps is strictly increasing.

1. **Path Definition:** Let e_i be the edge connecting v_{i-1} to v_i . Let $H(e_i) = t_i$.
2. **Adjacency Relation:** For any step i where $1 \leq i < k$: The edge e_i terminates at v_i . Therefore, $e_i \in \text{In}(v_i)$. The edge e_{i+1} originates at v_i .
3. **Application of Assignment Rule:** The timestamp t_{i+1} for edge e_{i+1} is calculated relative to $\text{In}(v_i)$.

$$t_{i+1} = 1 + \max(\{H(k) \mid k \in \text{In}(v_i)\} \cup \{0\})$$

4. **Inequality Derivation:** Since $e_i \in \text{In}(v_i)$, it follows that:

$$\max(\{H(k) \mid k \in \text{In}(v_i)\}) \geq H(e_i) = t_i$$

Substituting this into the assignment rule:

$$t_{i+1} \geq 1 + t_i$$

$$t_{i+1} > t_i$$

IV. Conclusion

The timestamp function H enforces a strict total ordering on all causal chains.

$$t_1 < t_2 < \dots < t_k$$

This monotonicity guarantees that the causal graph is a Directed Acyclic Graph (DAG), as any cycle would require the contradiction $t_i < t_i$.

Q.E.D.

1.3.Z Implications and Synthesis

Causal Graph

A network of relations has replaced the coordinate system. The timestamp functions as a permanent label. It freezes the moment of creation for every link and embeds the arrow of time directly into the topology. This creates a static skeleton. It is a record of events and their causes that stands independent of any observer. We have successfully translated the abstract concept of causality into a concrete, countable structure. This graph is the absolute floor of reality. Beneath this graph there is no sub-structure. There is only the logic of the code itself.

This structure provides the memory of the system. It encodes the past interactions that define the present state. In this ontology, space is not a pre-existing container that events happen within. Space is the relationship between events. If two particles are far apart, it is not because there is a lot of empty void separating them. It is because the graph distance is large. The graph distance is the sheer number of causal links one must traverse to get from one to the other. This is a background-independent description of reality that does not require an external ruler or grid. By embedding the timestamp t_L onto the edges, we ensure that the graph is not just a spatial web. It is a spacetime history. It is a growing block of causal connections where the past is preserved in the topology of the present.

Our inquiry now turns to the dynamics. Defining the specific operations allowed to transform this graph from one moment to the next is the next logical step. We have the object, but we do not yet have the motion. We must determine how this static web becomes a living, evolving universe. A graph that sits eternally unchanged is not a physics. It is a painting. To breathe life into this structure, we must define the legal moves that can alter it. This leads us to the definition of the task space.

1.4 Task Space

Operations on the graph cannot be arbitrary because they must be rigidly constrained by physical necessity. If we allowed the substrate to mutate without restriction, we would find ourselves in a universe with infinite degrees of freedom. This would lack the continuity required for the emergence of persistent physical laws. We are therefore compelled to identify the absolute minimum set of operations capable of transforming one state into another while preserving the discrete integrity of the events themselves. We cannot simply allow nodes to appear or disappear at random. The transformation must be continuous and conservative to maintain the coherence of physical objects over time.

Our investigation explores the fundamental symmetry between the act of forging a connection and the act of severing it. We seek a balance that permits the universe to breathe by expanding and contracting its relational web without requiring an external architect to direct every change. We must find a mechanism that allows complexity to arise from simplicity. We must use only local operations that do not require knowledge of the global state. This constraint ensures that physics remains local and causal. It prevents

“spooky action at a distance” from being baked into the fundamental rules. The mechanism must be blind to the whole, acting only on the immediate neighborhood.

Our inquiry restricts the domain of admissible transformations to a Task Space containing only those moves that are kinematically possible. We find that a vast repertoire of complex actions is unnecessary because a minimal set of primitive operations suffices to describe all possible evolutions. We distinguish between the additive process, which increases the relational density, and the subtractive process, which prunes it. These operations stand as inverses to one another. This ensures that the fundamental dynamics remain reversible in principle, even if the statistical behavior of the system eventually renders them irreversible in practice.

1.4.1 Definition: Elementary Task Space

Delimitation of Admissible Transformations by Kinematic Constraints

$\mathfrak{T}$ comprises the set of all graph transformations on the causal graph substrate $G = (V, E, H)$:

$$\mathfrak{T} = \{T : G \rightarrow G' \mid G' \text{ preserves acyclicity, monotonicity of } H, \text{ and finite cardinality}\}.$$

Each task $T \in \mathfrak{T}$ specifies an abstract input-output mapping: $\{\text{Input Attribute} \rightarrow \text{Output Attribute}\}$, where attributes denote isomorphism classes of subgraphs (e.g., the presence or absence of a directed edge $e = (u, v)$). Kinematic possibility here signifies structural admissibility: transformations must not invoke infinite resources, permit retroactive revisions to timestamps, or violate the irreflexive causal primitive defined by **Directed Causal Link**. The preservation of acyclicity ensures that G' admits no directed cycles (enforcing **Acyclic Effective Causality**), monotonicity of H requires that new timestamps exceed predecessors under **Monotonicity of History**, and finite cardinality bounds $|V'| \leq |V| + k$ for constant k (preventing unbounded blooms). Independent of probabilistic weighting or energetic viability, $\mathfrak{T}$ enumerates exhaustively “what can be built” from the discrete relations, serving as the kinematic substrate upon which dynamical laws impose selection (Abramsky, 2023).

1.4.2 Postulate: Vacuum Repertoire

Restriction of the Vacuum Repertoire to Primitive Edge Operations due to Catalytic Reciprocity

The set of fundamental kinematic operations available to the Universal Constructor is restricted exclusively to the following primitives: 1. **Edge Addition** ($\mathfrak{T}_{add}$): The insertion of a directed edge (u, v) into E , subject to the monotonic timestamp assignment. 2. **Edge Deletion** ($\mathfrak{T}_{del}$): The removal of a directed edge (u, v) from E . The theory admits no primitives for the direct creation or destruction of vertices independent of edge topology; vertices emerge solely as the endpoints of relations.

The **Vacuum Repertoire** delimits the kinematic capabilities of the fundamental substrate to exactly two primitive operations. This restriction asserts that the unmediated vacuum possesses no intrinsic capacity for higher-order transformations; operations such as simultaneous multi-edge generation, non-local topological swaps, or geometric smoothing do not exist as fundamental primitives. Instead, the theory mandates that all complex structural evolution derives exclusively from the iterative composition of these binary edge fluxes. The ambient relational structure functions as the auto-catalyst for these operations, requiring no extrinsic constructor to drive the basal dynamics. By confining the repertoire to this symmetric duality, this postulate enforces an ontological neutrality, ensuring that physical laws emerge not from ad hoc kinematic privileges but as constraint-based filters acting upon a uniform combinatorial potential.

1.4.3 Commentary: Primitive Tasks

Symmetry of Edge Creation and Deletion as Fundamental Fluxes

In the architecture of Graph Rewriting Systems, the foundational primitive manifests as vertex substitution: the targeted replacement of a local subgraph motif via a rewrite rule $A \rightarrow B$, where A and B denote finite templates matched isomorphically within G . For Quantum Braid Dynamics, this primitive realizes exclusively through two symmetric tasks on E :

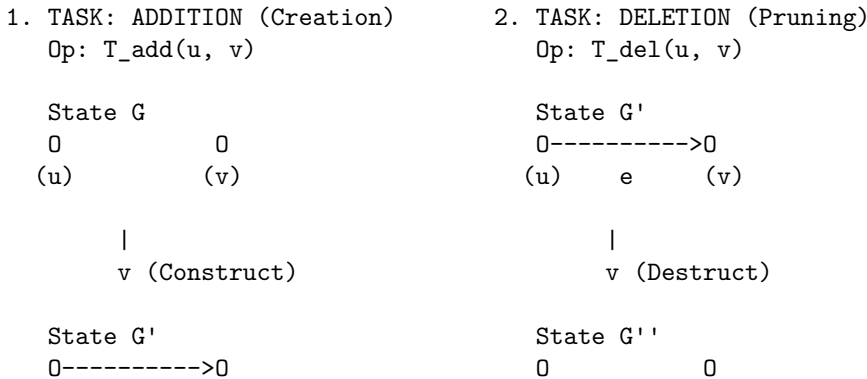
- $\mathfrak{T}_{add}$: The transformation $G \rightarrow G + e$, where $e = (u, v) \notin E$ and $u \neq v$, accretes the novel causal link with emergent timestamp $H(e) = t_L$ via the rewrite rule. This task instantiates a primitive causal relation, extending the relational horizon and enabling mediated influences (e.g., closing a compliant 2-path to nucleate a 3-cycle **Geometric Quantum**).
- $\mathfrak{T}_{del}$: The transformation $G \rightarrow G - e$, where $e = (u, v) \in E$, excises the link while preserving the historical imprint $H(e)$ and the acyclicity of G' . This task contracts superfluous connections, resolving topological tensions (e.g., pruning redundant paths to enforce parsimony under **The Deletion Probability**).

$\mathfrak{T}_{del}$ is defined as a topological modification, not an informational erasure. Within the Elementary Task Space, the excision of a causal link e removes the *active relation* (causal influence) but does not retroactively annihilate the *event of its creation*. The task space assumes an “Append-Only” metaphysics regarding the Global Sequencer’s log: t_L at which e was created remains a persistent property of the universe’s trajectory, even if the geometric constituent e is removed from the active graph G . This distinction allows for the pruning of geometry without the paradox of altering the past. Critically, this append-only historical poset $()$ incurs zero runtime memory overhead; when e is pruned by $\mathfrak{T}_{del}()$, it is fully excised from the active state graph data structure, maintaining strict structural sparsity and computational efficiency without retaining inactive or historically deleted edges in memory.

These primitives form the “assembly language” of $\mathfrak{T}$: every complex transformation, be it the braiding of fermionic worldlines, the curvature gradients of spacetime, or the entanglement webs of holography, decomposes into a countable sequence of such substitutions. Unlike general graph rewriting systems, where arbitrary motifs proliferate, Quantum Braid Dynamics restricts rewrite templates to these edge-level operations, ensuring that vertex identities remain purely relational and pre-geometric under the **Monotonicity of History** . The symmetry between creation and deletion reflects the reversibility constraint (Abramsky, Barbosa, & Searle, 2024) of Constructor Theory: if $\mathfrak{T}_{add}$ qualifies as possible (i.e., a constructor exists to enact it reliably), then its inverse $\mathfrak{T}_{del}$ must also qualify as possible, conserving the distinguishability of graph states without informational loss. This explicit duality mandates the equiprimordiality: the vacuum admits both fluxes symmetrically, with no primitive favoring one over the other, thereby embedding conservation of relational distinguishability at the ontological core.

1.4.3.1 Diagram: Task Repertoire

Depiction of Primitive Graph Fluxes via Addition and Deletion Operations



(u) e (v) (u) (v)

CONSTRAINTS:

1. Acyclicity: Addition cannot close a loop (unless 3-cycle).
 2. Monotonicity: $H(e) = \text{Current } t_L$.
 3. Reversibility: If Add is possible, Del is possible.
-

1.4.4 Commentary: Symmetry and Catalysis

Thermodynamic Reciprocity of Construction and Destruction under the Reversibility Constraint

The duality of $\mathfrak{T}_{add}$ and $\mathfrak{T}_{del}$ transcends mere convenience; it encodes the *catalytic reciprocity* of Constructor Theory, where creation and annihilation serve as thermodynamic conjugates in the ledger of relational becoming. This reciprocity grounds in Constructor Theory’s Reversibility Constraint, a foundational law of information conservation: if a task $T \in \mathfrak{T}$ qualifies as possible (i.e., a constructor exists to convert state A to state B reliably, with probability approaching 1 in the asymptotic limit), then the inverse task $B \rightarrow A$ must also qualify as possible, ensuring no physical process annihilates distinguishability without a reversible counterpart. In the causal graph, this constraint mandates the equiprimordiality of edge creation and deletion: $\mathfrak{T}_{add} : G \rightarrow G + e$ qualifies as admissible only if $\mathfrak{T}_{del} : G + e \rightarrow G$ remains viable, preserving isomorphism classes of graph states across the task space without informational erasure. Violations, such as irreversible mergers of vertices or phantom links persisting post-deletion, would render the substrate non-unitary, incompatible with the interoperability of quantum attributes in the extended framework. Thus, the Add/Del symmetry constitutes not an arbitrary postulate but a direct consequence of this constraint, elevating the graph’s mutability from combinatorial whim to a conserved relational currency, where each flux operation upholds the theory’s commitment to reversible possibility.

In the primordial vacuum, additions predominate, kindling quanta from relational sparsity akin to inflationary nucleation. In the equilibrated manifold, deletions enforce entropic bounds, sculpting cosmic voids without retroactive erasure of histories. This symmetry anticipates the master equation’s **Macroscopic Evolution** : net complexity accrues not from intrinsic bias but from the geometry of task densities, with the vacuum itself functioning as the universal catalyst (a persistent topological scaffold that facilitates substitutions while invariant under its own isomorphism class). Physically, this duality mirrors the Lagrangian’s dual gradients: ascent through addition, descent through deletion, tracing geodesics of minimal informational action across the task landscape. The substrate’s impartiality thus preserves: $\mathfrak{T}$ as neutral potential, awaiting the chiral adjudication of axioms and thermodynamic engines to impart directionality, much as parity violation selects helicity from symmetric braids in the fermionic sector.

1.4.5 Commentary: Task Independence

Independence of Kinematic Possibility from Dynamical Probability through Task Modularity

A defining virtue of this task-theoretic formulation resides in its kinematic purity: membership in $\mathfrak{T}$ invokes no oracle of probability, no calculus of free energy, nor any measure of dynamical preferability. The space enumerates merely the structural feasibility of flux, remaining agnostic to enactment frequency or energetic toll. An addition $\mathfrak{T}_{add}(u, v)$ qualifies if irreflexive and compliant with the **Monotonicity of History** , but its thermodynamic viability ($\Delta F < 0$ at vacuum temperature) defers to the **Addition Mode** . Deletions preserve H ’s monotonicity yet postpone Landauer costs until **Deletion Mode** is active.

This stratification upholds **Coherentist Justification** : ontology affords the task space, axioms constrain its potential to **Unique Causality (PUC)** , and dynamics impose the **Universal Constructor** . The vacuum’s relationality thus emerges as the agent of becoming: persistent yet enabling the full cycle of

construction that begets the universe from nullity. This independence ensures modularity: alterations to dynamical parameters (e.g., temperature scaling) perturb selection without reshaping kinematic possibility, facilitating isolation of ontology from mechanism and permitting the theory’s scalability across regimes.

1.4.Z Implications and Synthesis

Task Space

Limiting dynamics to the bare minimum allows the system simply to make or break a link. This symmetry reveals itself as a vital feature of the theory because it ensures the universe is not structurally biased by its own mechanics toward either infinite density or total emptiness. We have ensured that the machinery of the universe is neutral. This allows the outcome to be determined by the interaction of the parts rather than the design of the tools. This neutrality is essential. If the laws of physics were biased toward creation, the universe would explode instantly. If they were biased toward destruction, it would vanish.

Structures can be built and dissolved with equal facility. This allows the system to explore its configuration space freely. This neutrality guarantees that any order that eventually emerges does so because of the thermodynamic rules of selection, not because the kinematic machinery was predisposed to produce it. By restricting the universe to these two operations, we establish a conservation of possibility. Nothing is created that cannot be destroyed, and nothing is destroyed that cannot be recreated. This balance allows for a dynamic equilibrium to eventually form. It creates a state of flux that mimics the stability of matter.

This kinematic freedom is necessary but insufficient. While the ability to add and delete edges provides the vocabulary of change, it does not provide the grammar. A universe that can do anything at random will likely do nothing coherent. We have defined the verbs of our physical language, which are the creation and destruction of relations. However, we have not yet defined the sentences. We need to understand the vocabulary of shapes that these simple additions and deletions can form. We need to know which of those shapes represent valid physical structures versus mathematical noise. We turn now to the definition of the fundamental topological structures.

1.5 Graph-Theoretic Definitions

Extracting meaningful patterns from the noise of raw connectivity is our next logical task. A single link serves merely as a connection. When links chain together, they create higher-order topological meaning that we must learn to interpret. We cannot simply count edges because we must understand how they arrange themselves to form the fabric of geometry. We are looking for the emergent properties of the network that will eventually look like distance and area. We must define what it means to be “inside” or “outside” a structure that has no physical volume, relying purely on the topology of the connections.

We seek the smallest possible structure capable of enclosing a region of the graph, thereby defining the concept of an interior. It becomes necessary to distinguish between open chains, which transmit influence from one locus to another, and closed loops, which define self-reference and stability. We require a vocabulary to describe these shapes because they will eventually serve as the immutable atoms of our geometry. Without this classification, the graph remains a chaotic tangle without distinguishing features. It is a static noise that contains no information. We must learn to read the geometry hidden in the algebra.

Our analysis is confined to the most basic topological motifs to avoid premature complexity. We identify the unit of interaction as an open sequence allowing one event to reach another. This establishes the concept of transitivity without defining it via coordinates. We contrast this with the unit of stability, which we identify as the smallest possible loop. This is a structure that allows feedback without traversing a vast distance. We must also distinguish these stable forms from longer, more tenuous loops, which we will later find to be

dynamically unstable. This taxonomy provides the “periodic table” of graph elements from which we will construct the universe.

1.5.1 Definition: Fundamental Graph Structures

Classification of Allowable Topologies by Definitions of Acyclicity and Bipartiteness

The following structures constitute the vocabulary for topological constraints:

- **Directed Acyclic Graph (DAG):** A directed graph containing no directed cycles. A DAG represents a universe with a strict causal order, where it is impossible for an event to be its own cause (Diestel, 2017).
 - **Bipartite Graph:** A graph where the set of vertices V can be divided into two disjoint sets, V_A and V_B , such that every edge connects a vertex in V_A to one in V_B .
 - **Directed Path:** A sequence of vertices $(v_0, v_1, \dots, v_n)$ such that for all i , the directed edge $(v_i, v_{i+1}) \in E$.
 - **Simple Path:** A path containing no repeated vertices.
-

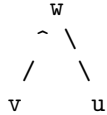
1.5.2 Definition: 2-Path

2-Path as the Minimal Unit of Transitive Mediation

A **2-Path** is defined as a simple Directed Path of length exactly 2, denoted as the ordered triplet (v, w, u) , such that $(v, w) \in E$ and $(w, u) \in E$. This structure constitutes the minimal unit of transitive mediation (Bondy & Murty, 2008) required for the rewrite rule to identify a potential closure site.

1.5.2.1 Diagram: Open 2-Path

Visualization of Transitive Mediation within the Open 2-Path Structure



1.5.3 Definition: Cycle Definitions

Distinction between Forbidden and Permitted Cyclic Structures through the Hierarchy of Cycle Lengths

A **Cycle** is defined as a non-trivial Directed Path $(v_0, \dots, v_k)$ where $v_0 = v_k$. 1. **2-Cycle:** A Cycle of length $k = 2$, representing immediate reciprocal causality between two events. 2. **3-Cycle:** A Cycle of length $k = 3$, representing the minimal closed loop enclosing a topological area (Janson, 1987) (the Geometric Quantum).

1.5.3.1 Diagram: Closed 3-Cycle

Comparison of Transitive Flow and Cyclic Closure through Topological Motifs

OPEN 2-PATH (Pre-Geometric)

"Correlation without Area"

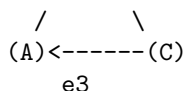
CLOSED 3-CYCLE (Geometric Quantum)

"The Smallest Area / Stable Bit"





Relation: $A \rightarrow B, B \rightarrow C$
 Status: Transitive Flow



Relation: $A \rightarrow B \rightarrow C \rightarrow A$
 Status: Self-Reference / Closure

1.5.Z Implications and Synthesis

Graph-Theoretic Definitions

Identification of the fundamental motifs gives us our building blocks for the chapters to come. The open path represents the potential for interaction and causal flow. The closed loop represents the realization of structure and geometric area. These simple shapes constitute the alphabet of our physical geometry. We are building the periodic table of graph elements. We are identifying the stable isotopes of connectivity that can endure in a fluctuating universe. Without these definitions, we would be unable to distinguish a random tangle from a meaningful structure like a particle or a vacuum manifold.

By defining them clearly, we give the system the capacity to recognize its own local topology. We distinguish between a connection and a closure. This is the first step toward the emergence of geometry from pure relation. An open path defines a one-dimensional causal relation, a sequence of before and after. A closed loop defines a two-dimensional area, a boundary that separates inside from outside. By categorizing these shapes, we prepare the ground for a physics that constructs dimensionality from the bottom up, rather than assuming it as a background stage. The graph is no longer just a list of edges. It is a collection of geometric objects waiting to be assembled into a manifold.

With the definitions in place, establishing the laws that dictate which of these shapes are permitted and which are forbidden is necessary. This leads directly to the constraints. We have the canvas and the paint, but we do not yet have the composition. We have assembled the complete ontological toolkit involving the iterator, the graph, the operations, and the shapes. But a toolkit is not a blueprint. We must now enact the laws that govern how these tools are used. We must ensure that the universe they build is consistent and causal. We turn to Chapter 2 to legislate the Axioms.

1.6 Formal Synthesis

End of Chapter 1

Avoiding the shifting sands of space and time, our inquiry anchors the universe in the bedrock of discrete events and causal links. By rejecting the siren call of the continuum, this chapter establishes a finite, computable substrate where “where” is defined strictly by connectivity and “when” by the relentless iterator t_L . The graph is a living record of existence, growing step by step from a definitive origin.

Yet, raw potential is indistinguishable from chaos; without legislation, the graph risks tangling into circular logic or fragmenting into disjoint realities. The urgent need for constraints becomes clear: a physical universe must be more than mathematically possible, it must be causally coherent. The infinite degrees of freedom require pruning to ensure history remains linear and logic remains sound.

The *substance* of reality is now established, but its *laws* remain unwritten. To carve a cosmos out of this raw potential, strict axioms must distinguish the physically valid from the merely constructible. We turn now to **Chapter 2**, where the fundamental rules of existence will be enacted.

Table of Symbols

Symbol	Description	Context / First Used
$\mathfrak{S}$	A finite formal system	Sec.1.1.1
$\mathcal{A}$	The Axiomatic Basis (set of foundational postulates)	Sec.1.1.1
$\mathfrak{D}$	A Formal Deductive System tuple $(\mathcal{L}, \mathcal{A}, \mathcal{I})$	Sec.1.1.2
$\mathcal{L}$	The Formal Language (alphabet and grammar)	Sec.1.1.2
$\mathcal{I}$	The set of Rules of Inference	Sec.1.1.2
$\vdash$	Syntactic derivability (provability)	Sec.1.1.2
$\models$	Semantic entailment (truth)	Sec.1.1.2
Γ	A set of premises	Sec.1.1.2
θ	A derived theorem	Sec.1.1.2
$\mathfrak{F}$	A consistent system capable of primitive recursive arithmetic	Sec.1.1.3
$\mathcal{G}$	The Gödel sentence (true but unprovable)	Sec.1.1.3
$Con(\mathfrak{F})$	The consistency statement of system $\mathfrak{F}$	Sec.1.1.3
$\perp$	Logical contradiction	Sec.1.1.6
t_L	Global Logical Time (discrete iteration counter)	Sec.1.2.1
t_{phys}	Physical Time (emergent, geometric)	Sec.1.2.1
$\mathbb{N}_0$	Set of non-negative integers (Domain of t_L)	Sec.1.2.1
U_{t_L}	Global state of the universe at step t_L	Sec.1.2.2
$\mathcal{U}$	Universal Evolution Operator	Sec.1.2.2
$\hat{H}$	Hamiltonian constraint operator	Sec.1.2.2
Ψ	Wavefunction of the universe	Sec.1.2.2
τ	Fictitious time parameter (Stochastic Quantization)	Sec.1.2.2.1
μ	Renormalization scale	Sec.1.2.2.1
$\hat{P}$	Permutation Operator (CAI interpretation)	Sec.1.2.2.2
$\mathcal{T}$	Unimodular Time variable	Sec.1.2.2.3
$\Lambda, \hat{\Lambda}$	Cosmological Constant (variable/operator)	Sec.1.2.2.3
$S(U)$	Information content/Entropy of state U	Sec.1.2.3
$\mathcal{O}(\cdot)$	Big O notation (asymptotic growth)	Sec.1.2.3
Ω_t	Set of admissible physical states at time t	Sec.1.2.3.1
b	Finite Branching factor	Sec.1.2.3.1
s_t	Surface area (active degrees of freedom)	Sec.1.2.3.1
δ_{holo}	Holographic scaling constant	Sec.1.2.3.1

Symbol	Description	Context / First Used
T	Temporal Domain (Set of integers)	Sec.1.2.4.1
$\mathbb{Z}_{\leq 0}$	Set of non-positive integers (Infinite Past domain)	Sec.1.2.4.1
$\mathcal{H}_{\text{hist}}$	History sequence (set of operations)	Sec.1.2.4.1
μ	Mean of entropy production (Context: Statistics)	Sec.1.2.4.1
σ^2	Variance of entropy production	Sec.1.2.4.1
ΔI_k	Information bit contribution	Sec.1.2.4.1
Ω	Universal State Space (Set of all admissible graphs)	Sec.1.2.5.1
$\mathcal{P}_T$	Trajectory sequence (Context: Recurrence Proof)	Sec.1.2.5.1
$\prec$	Strict causal precedence	Sec.1.2.5.1
$\epsilon(op)$	Energy cost per operation	Sec.1.2.6.1
E_{total}	Total energy dissipated	Sec.1.2.6.1
k_B	Boltzmann constant	Sec.1.2.6.2
T	Temperature (Context: Thermodynamics)	Sec.1.2.6.2
$\hbar$	Reduced Planck constant	Sec.1.2.6.2
c	Speed of light	Sec.1.2.6.2
$G_{\mu\nu}$	Einstein Tensor	Sec.1.2.6.2
$T_{\mu\nu}$	Continuous stress-energy tensor	Sec.1.2.6.2
R_s	Schwarzschild Radius	Sec.1.2.6.2
U_0	The unique initial state	Sec.1.2.7
R_n	The n -th Grim Reaper entity	Sec.1.2.7.2
G	A specific Causal Graph (V, E, H)	Sec.1.3.1
V	Set of Vertices (Abstract Events)	Sec.1.3.1
E	Set of Directed Edges (Causal Relations)	Sec.1.3.1
H	History Function (Timestamp map $E \rightarrow \mathbb{N}$)	Sec.1.3.1
v, u, w	Individual vertices	Sec.1.3.1
e	Individual edge (u, v)	Sec.1.3.1
$\text{In}(u)$	Set of incoming edges to vertex u	Sec.1.3.4.1
$\mathfrak{T}$	Elementary Task Space	Sec.1.4.1
$\mathfrak{T}_{\text{add}}$	Primitive Task: Edge Addition	Sec.1.4.2
$\mathfrak{T}_{\text{del}}$	Primitive Task: Edge Deletion	Sec.1.4.2
ΔF	Change in Free Energy	Sec.1.4.5
V_A, V_B	Disjoint vertex partitions (Bipartite definition)	Sec.1.5.1

References

- **Anderson, E. (2012).** *The Problem of Time in Quantum Gravity*. Annalen der Physik, 524(12), 757-786. Available at: <https://arxiv.org/abs/1009.2157>
- **Bennett, C. H. (1982).** *The thermodynamics of computation-a review*. International Journal of Theoretical Physics, 21(12), 905-940. Available at: <https://link.springer.com/article/10.1007/BF02084158>
- **Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987).** *Space-time as a causal set*. Physical Review Letters, 59(5), 521. Available at: <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.59.521>
- **Bondy, J. A., & Murty, U. S. R. (2008).** *Graph Theory*. Springer. Available at: <https://link.springer.com/book/9781846289699>
- **Diestel, R. (2017).** *Graph Theory (5th ed.)*. Springer. Available at: <https://diestel-graph-theory.com/>
- **Enderton, H. B. (2001).** *A Mathematical Introduction to Logic (2nd ed.)*. Academic Press. Available at: <https://www.sciencedirect.com/book/9780122384523/a-mathematical-introduction-to-logic>
- **Gambini, R., García-Pintos, L. P., & Pullin, J. (2023).** *The Page-Wootters mechanism in canonical quantum gravity*. Classical and Quantum Gravity, 40(12), 125008. Available at: <https://arxiv.org/abs/0809.4235>
- **Gödel, K. (1931).** *On Formally Undecidable Propositions of Principia Mathematica and Related Systems*. Monatshefte für Mathematik und Physik, 38, 173-198. Available at: https://homepages.uc.edu/~martinj/History_of_Logic/Godel/Godel%20%E2%80%93%20On%20Formally%20Undecidable%20Propositions%20of%20Principia%20Mathematica%201931.pdf
- **Janson, S. (1987).** *Poisson approximation for large cycles in random graphs*. Random Structures & Algorithms, 2(3). Available at: <http://stat.wharton.upenn.edu/~steele/Courses/531/531Resources/Janson1987PoissonProcess.pdf>
- **Lamport, L. (1978).** *Time, clocks, and the ordering of events in a distributed system*. Communications of the ACM, 21(7), 558-565. Available at: <https://doi.org/10.1145/359545.359563>
- **Landauer, R. (1991).** *Information is Physical*. Physics Today, 44(5), 23-29. Available at: <https://doi.org/10.1063/1.881299>
- **Marker, D. (2002).** *Model Theory: An Introduction*. Springer. Available at: <https://link.springer.com/book/10.1007/b98860>
- **Page, D. N., & Wootters, W. K. (1983).** *Evolution without evolution: Dynamics described by stationary observables*. Physical Review D, 27(12), 2885. Available at: <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.27.2885>
- **Palmigiano, A., & Sadrzadeh, M. (Eds.). (2023).** *Samson Abramsky on Logic and Structure in Computer Science and Beyond*. Springer. Available at: <https://link.springer.com/book/10.1007/978-3-031-24117-8>
- **Rovelli, C. (1996).** *Relational Quantum Mechanics*. International Journal of Theoretical Physics, 35(8), 1637-1678. Available at: <https://arxiv.org/abs/quant-ph/9609002>
- **Rovelli, C., & Smolin, L. (1990).** *Loop space representation of quantum general relativity*. Nuclear Physics B, 331(1), 80-152. Available at: [https://doi.org/10.1016/0550-3213\(90\)90019-A](https://doi.org/10.1016/0550-3213(90)90019-A)
- **Sorkin, R. D. (2005).** *Causal sets: Discrete gravity*. Lectures on Quantum Gravity (pp. 305-327). Springer. Available at: <https://arxiv.org/abs/gr-qc/0309009>
- **Wolfram, S. (2002).** *A New Kind of Science*. Wolfram Media. Available at: <https://www.wolframscience.com/nks/>
- **Wolfram, S. (2020).** *A Project to Find the Fundamental Theory of Physics*. Wolfram Writings. Available at: <https://writings.stephenwolfram.com/2020/04/finally-we-may-have-a-path-to-the-fundamental-theory-of-physics-and-its-beautiful/>
- **Abramsky, S., Barbosa, R. S., & Searle, A. (2024).** *Combining contextuality and causality: a game semantics approach*. Philosophical Transactions of the Royal Society A, 382(2268), 20230002. Available at: <https://royalsocietypublishing.org/doi/10.1098/rsta.2023.0002>

Document Status

Draft Version 0.2

DOI: 10.5281/zenodo.18124967

Copyright © 2025 Braid Dynamics. All Rights Reserved. This document is provided for personal, educational, and academic research purposes only. Dissemination, reproduction, or commercial use is strictly forbidden without prior written permission from the author.